

Engineering Economics bata padhi.

Date _____
Page _____

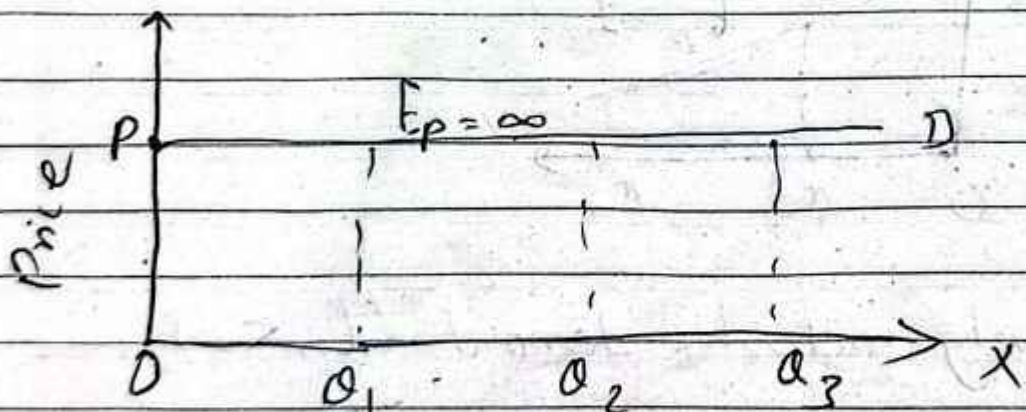
- Elasticity of demand $\rightarrow$ law of demand par lekhna parna gu
change in price $\rightarrow$ change in demand
change in income $\rightarrow$ change in demand.

- E_p = - Percentage change in quantity demanded
percentage change in price

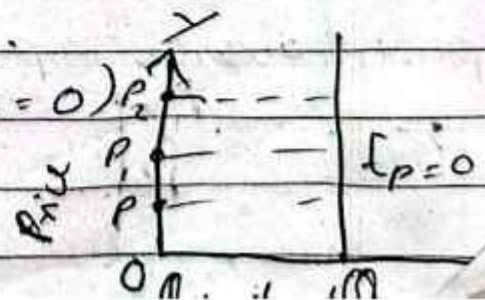
$$\therefore E_p = - \frac{\Delta Q}{\Delta P} \cdot \frac{P}{Q}$$

- Types of price elasticity of demand

- Perfectly elastic demand ($E_p = \infty$)
price $\rightarrow$ no change
demand fluctuate

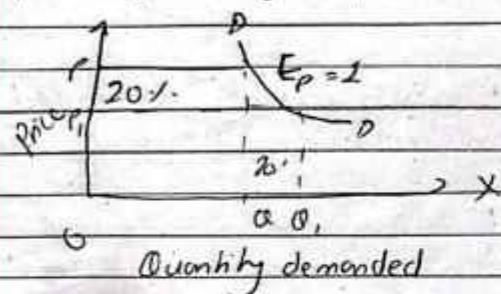


- Perfectly inelastic demand ($E_p = 0$)
 $E_p = 0$
any amount



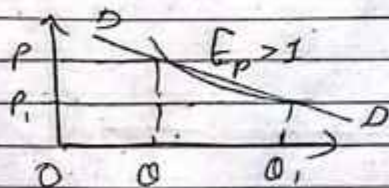
- Unitary Elastic demand ($E_p = 1$)

percentage change in price = percentage change in demand



- Relatively elastic demand ($E_p > 1$)

(20%) (40%)
change in price < change in demand



- Relatively inelastic demand ($E_p < 1$)

percent change in price > percentage change in demand.



(*) Price of commodity X decreases from 10 to 8

initial price (P) = 10

New price (P_1) = 8

change in price = $P_1 - P = 8 - 10 = -2$

change in quantity = $Q_1 - Q = 60 - 50 = 10$

coefficient of price of elasticity (E_p) = 2

$$E_p = - \frac{\Delta Q}{\Delta P} \times \frac{P}{Q} = - \frac{10}{-2} \times \frac{10}{50} = 1$$

(ii) Income elasticity of demand:-

$$E_y = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

• Types:

- Positive income elasticity of demand ($E_y > 0$)
- Income elasticity greater than unity ($E_y > 1$)
- Income elasticity equals to unity ($E_y = 1$)
- Income elasticity less than unity ($E_y < 1$)
- Zero income elasticity ($E_y = 0$)

- Negative income elasticity < 0

Eg
-x

When customer income rises from Rs 100 to Rs 102 his demand for goods X increases from 25 units per week to 30 units per week. Then find out the elasticity of demand for goods X.

initial income (Y) = 100
new income (Y_1) = 102
initial demand (Q) = 25 unit
new demand (Q_1) = 30 unit

$E_y = ?$
we have

$$E_y = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

$$Y = Y_1 - Y = 102 - 100 = 2$$

$$Q = Q_1 - Q = 30 - 25 = 5 \text{ unit}$$

$$E_y = \frac{5}{2} \times \frac{100}{25}$$

$$= 10$$

Date _____
Page _____

(*) Cross elasticity of demand E_c

$E_{xy} = \frac{\text{Percentage change in demand for good-X}}{\text{Percentage change in price of good-Y}}$

$$= \frac{\Delta Q_x}{\Delta P_y} \times \frac{P_y}{Q_x}$$

(*) Positive cross elasticity of demand ($E_c = +ve$ or ($E_{xy} > 0$))

goods n price $\uparrow$ leads to goods x demand $\uparrow$

(*) Negative cross elasticity of demand

(*) Zero cross elasticity of demand $E_c = 0$

(*) Calc the cross elasticity between Samosa and Pakanda using following table.

Soln

$$\Delta Q_n = 2 - 1 = 1$$

$$\Delta P_y = 15 - 10 = 5$$

$$Q_n = 1$$

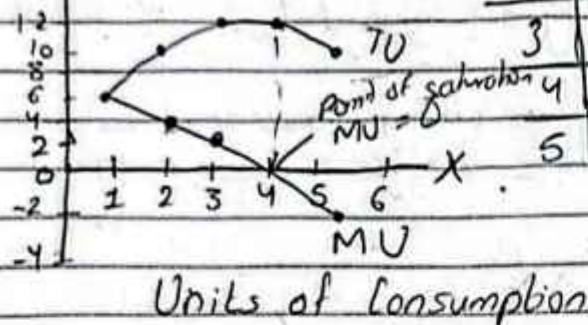
$$P_y = 10$$

$$E_{xy} = \frac{1}{5} \times \frac{10}{1} = 2 \text{ Ans}$$

Date _____
Page _____

Demand creates its own supply

TU/MU



UOC	TU	MU
1	6	6
2	10	4
3	12	2
4	12	0
5	10	-2

Engineering Economics $\rightarrow$ Imp

1. Imp

- 7th principle of Engineering Economics

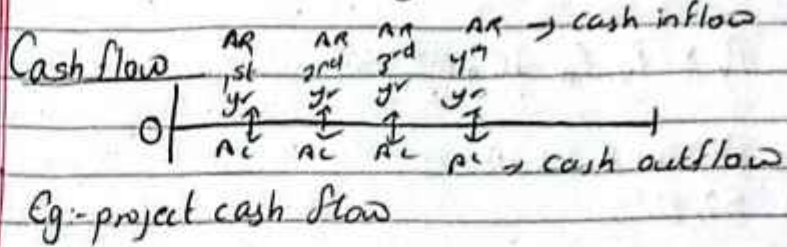
1. Develop the alternatives (remember quality does from quantity)
2. Focus on the differences
3. Use a consistent view point

AR - annual revenue $P_v \rightarrow$ Present value
AC - Annual cost.

Date _____
Page _____

Imp

(*) The time value of money.



Eg - project cash flow

(*) Techniques of time value of money:

(1) Future value / Future Worth (FV/FW)

$$F_v = P_v (1+i)^n \quad \dots (1)$$

F_v - Future value
 P_v - Present value
 i - interest rate
 n - no. of yrs

Example

Mr. X deposits Rs. 50,000 now in a bank account, how much money will be recd at the end of 5th year if interest rate is 10%.

$$\begin{aligned}
 P_v &= 50,000 \\
 i &= 10\% = 0.1 \\
 n &= 5 \text{ yrs} \\
 F_v &= ?
 \end{aligned}$$

we have

$$F_v = P_v (1+i)^n = 50,000 (1.1)^5 = 80,525.5$$

Fvif = future value interest factor

Date _____
Page _____

Alternatively:

$$F_v = P_v \times F_{vif} \text{ @ } 10\% \text{ 5 yrs}$$

$$= 50,000 \times 1.6105$$

$$= 80,525$$

(*) Calculate future value / Present value / Present worth (Pv/Pw)
 $P_v = 10,000$
 $i = 10\%$
 $n = 3 \text{ yrs}$

$$F_v = P_v \times F_{vif} \text{ @ } 10\% \text{ 3 yrs}$$

$$= 10,000 \times 1.3310$$

$$= 13,310$$

$$P_v = \frac{F_v}{(1+i)^n}$$

$$P_v = F_v (1+i)^{-n}$$

Alternatively:

Using table

$$P_v = F_v \times P_{vif} \text{ @ } i\% \text{ n yrs}$$

	11000	12100	13310
Pv	10000	11000	12100
	1	2	3

Note: Process of transferring P_v into F_v is called compounding.
 Process of converting F_v into P_v is called discounting.

Date _____
Page _____

JIMP

* Eg: Mr hari needs Rs. 5 Lakhs at the end of 20th year for his daughter's education. How much does he have to deposit now, if interest rate is 12%.

* How much will it take to double your investment if interest rate is 10%.

$$F_v = 2P_v$$

$$i = 10\% = 0.10$$

$$n = ?$$

Solⁿ

$$F_v = 5 \text{ Lakhs}$$

$$n = 20 \text{ yrs}$$

$$P_v = ?$$

$$i = 12\% = 0.12$$

Now

$$P_v = F_v (1+i)^{-n}$$

$$P_v = 2P_v (1+0.1)^{-n}$$

$$\text{on } 0.5 = (1+0.1)^{-n}$$

Now

$$P_v = F_v (1+i)^{-n}$$

$$= 5,00,000 (1+0.12)^{-20}$$

$$= 1,60,986.6183$$

$$\text{on } 0.5 = (1.1)^{-n}$$

$$\text{on } \frac{0.5}{\log(0.5)} = \frac{1}{\log(1.1)}$$

Alternatively

$$P_v = F_v \times P_{vif} \text{ @ } 12\% \text{ 20 yrs}$$

$$= 5,00,000 \times 3.1058$$

$$= 1,55,290$$

$$\text{on } 0.5 = (1.1)^{-n}$$

$$\text{on } -n = \log_{1.1}(0.5)$$

$$\text{on } n = 7.27$$

$$\therefore n = 7.27$$

Every year $\rightarrow$ 20,000 $\rightarrow$ deposited $\Rightarrow$ annuity.

Date _____
Page _____

(*) Annuity:

Conditions:-

(*) Every year deposit $\rightarrow$

- Same amount
- And amount deposited at end of the year.

$\rightarrow$ Every year amount

$$(i) F_v = A \left[\frac{(1+i)^n - 1}{i} \right] \quad \text{--- (ii)}$$

Alternative (using table)

$$F_v = A \times F_{v, i, n} \times i \% \times n \text{ yrs}$$

Example:-

- Mr. Y deposits Rs. 10,000 every year (annuity) in a bank account. How much money will he get at the end of 5th year, if interest rate is 10%.

Soln

$$\text{Annuity (A)} = \text{Rs } 10,000$$

$$F_v = ?$$

$$i = 10\% = 0.1$$

$$n = 5 \text{ yrs}$$

Now

Using table

Date _____
Page _____

$$F_v = A \left[\frac{(1+i)^n - 1}{i} \right]$$

$$= 10,000 \left[\frac{(1+0.1)^5 - 1}{0.1} \right]$$

$$= 61,051$$

$$F_v = A \times F_{v, i, n} \times i \% \times n \text{ yrs}$$

$$= 10,000 \times 6.1051$$

$$= 61,051$$

(*) Present value of Annuity:

$$P_v = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

Alternative

$$P_v = A \times P_{v, i, n} \times i \% \times n \text{ yrs}$$

- (*) Mr. B needs Rs 25,000 every year for 6 yrs. How much money does he have to deposit now to meet this requirement if interest rate is 12%.

Soln

Given,

$$\text{Annuity (A)} = \text{Rs } 25,000$$

$$n = 6 \text{ yrs}$$

$$i = 12\% = 0.12$$

Now

$$P_v = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$= 25,000 \left[\frac{(1+0.12)^6 - 1}{0.12(1+0.12)^6} \right] = 102,785.1831$$

Alternatively

$$P_v = A \times P_{v, i, n} \times i \% \times n \text{ yrs}$$

$$= 25,000 \times 4.1114$$

$$= 102,785$$

- ⑧ Z needs Rs 5,00,000 at the end of 10th year. How much money does he need to deposit every year if interest rate is 9%.

Now $FV = Rs\ 5,00,000$, $n = 10$, $i = 9\% = 0.09$
 $A = ?$

Now
$$FV = A \left[\frac{(1+i)^n - 1}{i} \right]$$

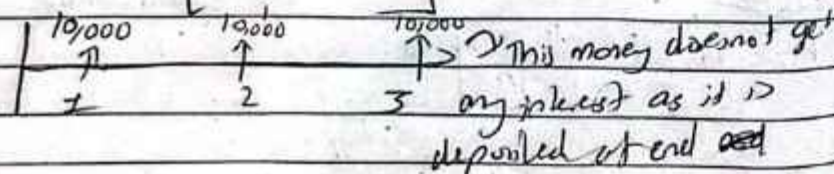
$$5,00,000 = A \left[\frac{(1+0.09)^{10} - 1}{0.09} \right]$$

$$5,00,000 = A \times 15.1929$$

$$A = 32910$$

- ⑨ Present value of Annuity

$$PV = A \left[\frac{(1+i)^n - 1}{i} \right]$$



Proof: Money deposited at last year don't get interest

Proof of formula. $FV = A \left[\frac{(1+i)^n - 1}{i} \right]$

- ⑩ Example

$FV = ?$, $n = 3$ yrs, $i = 10\% = 0.1$, $A = 10,000$

$$FV = A \times FVIFA, 10\% - 3$$

$$= 10,000 \times 3.3100$$

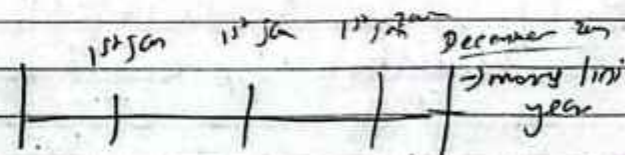
$$= 33,100$$

Diagram illustrating cash flows for an annuity due. A timeline from year 1 to year 3. At year 1, there is an upward arrow labeled 10,000. At year 2, there is an upward arrow labeled 10,000. At year 3, there is an upward arrow labeled 10,000. A note says "if".

Year 1 get $(n-1)$ yrs interest

$$= 10,000 \times (1.1)^2 + 10,000 \times (1.1)^1 + 10,000 \times (1.1)^0$$

$$= 33,100$$



- ⑪ Annuity due : in question
 → Depositing amount (at the beginning of the year) ^{due}
 → last year interest

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right] \times (1+i)$$

- ⑫ Example

How much money will Shyan receive at the end of 8 yrs if he deposits Rs 10,000 at

due.

the beginning of each year if interest rate is 8%.

Solⁿ

$n = 8 \text{ yrs}$, $A = \text{Rs } 10,000$, $i = 8\% = 0.08$

Since amount is deposited at beginning of every year this is case of annuity due.

$$F_v = A \left[\frac{(1+i)^n - 1}{i} \right] \times (1+i)$$

$$= 10,000 \left[\frac{(1+0.08)^8 - 1}{0.08} \right] (1+0.08)$$

$$= \text{Rs } 1,14,875$$

Imp

(*) Calculation of present value for uneven cash flow

Example:

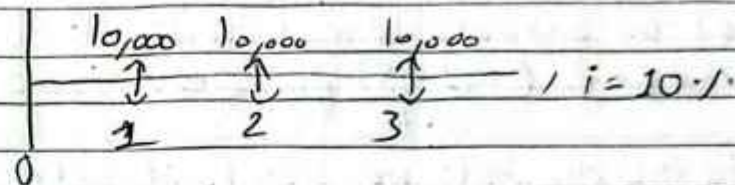
$A = 10,000$, $i = 10\%$, $n = 3 \text{ yrs}$, $P_v = ?$

$$P_v = A \times P_{vif} \times 10\% \times 3$$

$$= 10,000 \times 2.4869$$

$$= 24,869$$

Justification:



$$P_v = \frac{10,000}{(1.1)^1} + \frac{10,000}{(1.1)^2} + \frac{10,000}{(1.1)^3}$$

$$= 24,868.52$$

(*) Example

and FV

Gulai agave
P_v nikatni

$$P_v \text{ mi, } F_v = P_v \times (1+i)^n$$

Calculate P_v from the following cash flow sh

Year	1	2	3	4	5	6	
CF	5000	8000	3000	2000	1000	6000	$i = 10\%$

yeuagave

$$P_v = \frac{5000}{1.1} + \frac{8000}{(1.1)^2} + \frac{3000}{(1.1)^3} + \frac{2000}{(1.1)^4} + \frac{1000}{(1.1)^5} + \frac{6000}{(1.1)^6}$$

$$= 18,784.76$$

$$F_v = 5000 \times (1.1)^5 + 8000 \times (1.1)^4 + 3000 \times (1.1)^3 + 2000 \times (1.1)^2 + 1000 \times (1.1)^1 + 6000 \times (1.1)^0$$

$$= 33,278.35$$

- (*) Pemba was student at the university in 2007, she borrowed Rs 12,000 ~~at~~ student's loan at an annual interest rate of 9%. If she repays Rs 1488 per year, how long to the nearest year, will it take her to repay the loan?

Sol
long to the nearest year, will it take her to repay the loan?

$$P_v = \text{Rs } 12000$$

$$i = 9\% = 0.09$$

$$A = \text{Rs } 1488$$

$$n = ?$$

We know

$$P_v = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$12000 = 1488 \left[\frac{(1+0.09)^n - 1}{0.09(1+0.09)^n} \right]$$

$$8.0645 = \frac{(1.09)^n - 1}{0.09(1+0.09)^n}$$

$$0.7258 = (1+0.09)^n - 1$$

$$1 = (1.09)^n - 0.7258 \quad (1.09)^n$$

$$1 = 0.2742 (1.09)^n$$

$$3.6469 = (1.09)^n \Rightarrow n = \log_{1.09}(3.6469)$$

$$(1.09)^{15} = (1.09)^n \therefore n = 15 \text{ yrs} = 15.01 \text{ yrs}$$



- (*) Types of interest rate

- (i) Simple interest rate

Example

$$P_v = 100000$$

$$i = 10\%$$

$$n = 5 \text{ yrs}$$

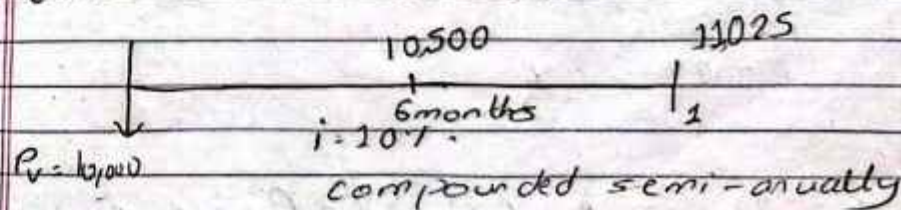
$$F_v = ?$$

$$\begin{aligned} F_v &= 100000 \times 0.1 \times 5 + 100000 \\ &= 50000 + 100000 \\ &= 150000 \end{aligned}$$

- (*) Compound interest

$$F_v = P_v (1+i)^n$$

- (ii) Effective interest rate (r)



$$\text{Interest rate} = \frac{1025}{10000} \times 100 = 10.25\% = r \text{ (Effective interest rate)}$$

Effective interest rate $r = \left(1 + \frac{i}{m}\right)^m - 1$

where,

m = no. of compounding in a year.

$$= \left(1 + \frac{0.1}{2}\right)^2 - 1 = 0.1025$$

$$= 10.25\%$$

Ex Example: Mr. Hari deposits Rs. 2,00,000 now in the bank account. How much money will he get at end of 8th year with 12% interest rate.

a) Following S.I. rate

b) Following annually compounding.

c) " ~~quarterly~~ " "

Sol

Given,

$$P_v = \text{Rs } 2,00,000$$

$$n = 8$$

$$i = 12\% = 0.12$$

$$F_v = ?$$

$$\begin{aligned} a) F_v &= 2,00,000 \times 0.12 \times 8 + P_v \\ &= 3,92,000 \end{aligned}$$

b) following compounding annually.

$$\begin{aligned} F_v &= P_v (1+i)^n \\ &= 2,00,000 (1+0.12)^8 \\ &= 4,95,192.6353 \end{aligned}$$

c) following compounding quarterly.

Since the interest rate

$$\text{Effective interest rate } (r) = \left(1 + \frac{i}{m}\right)^m - 1$$

Since interest is calculated on the basis of quarterly. i.e. calculate effective interest rate

$$\begin{aligned} r &= \left(1 + \frac{0.12}{4}\right)^4 - 1 \\ &= 12.55\% \end{aligned}$$

Now

$$\begin{aligned} F_v &= P_v (1+r)^n \\ &= 2,00,000 (1+0.1255)^8 \\ &= 5,14,984.30 \end{aligned}$$

Alternatively

→ ~~Ann~~ in case of $n \rightarrow n \times m \rightarrow (i)$
→ in case of $i \rightarrow i/m \rightarrow (i)$

$$FV = P_v (1+i)^n$$

$$= P_v (1+i_m)^{n \times m}$$

$$= 20000 \left(1 + \frac{0.12}{4}\right)^{4 \times 5}$$

$$= 5,15,016$$

(*) $\frac{EMI}{P_v}$ (monthly basis)

$$\text{Loan amount} = A \times \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right]$$

$$\checkmark \text{ Loan amount} = EMI \times \left[\frac{(1+i_m)^{n \times m} - 1}{i_m (1+i_m)^{n \times m}} \right]$$

(*) Mr. Poudel is willing to buy a new car costing at Rs. 50,00,000. He applied for a loan at NJMB Ltd. The bank asked him to arrange 20% cash as down payment. How much money does Mr. Poudel need to pay every month (i.e. EMI) for 5 yrs to settle the loan if interest rate is 8%.

m:12

I.A.

$$= 50,00,000$$

As down payment is 20%, so,

$$(\text{Loan amount}) P.v. = 40,00,000 = \text{loan amount}$$

$$n = 5 \text{ yrs}$$

$$m = 12$$

$$i = 8\% = 0.08$$

Now

$$\text{Loan amount} = EMI \times \left[\frac{(1+i_m)^{n \times m} - 1}{i_m (1+i_m)^{n \times m}} \right]$$

$$\therefore EMI = \text{Loan amount} \times \left[\frac{i_m (1+i_m)^{n \times m}}{(1+i_m)^{n \times m} - 1} \right]$$

$$= 40,00,000 \times \left[\frac{\frac{0.08}{12} \left(1 + \frac{0.08}{12}\right)^{5 \times 12}}{\left(1 + \frac{0.08}{12}\right)^{5 \times 12} - 1} \right]$$

$$= 81,105.57 \text{ Rs}$$

Alternatively:

By using monthly effective interest rate we have

$$i_{\text{monthly}} = (1+i)^{\frac{1}{12}} - 1$$

$$= (1.08)^{\frac{1}{12}} - 1$$

$$= 0.00643$$

$$EMI = \text{loan amount} \left[\frac{r(1+i)^n}{(1+i)^n - 1} \right]$$

$$= 40,00,000 \left[\frac{0.00643(1+0.00643)^5}{(1+0.00643)^5 - 1} \right]$$

$$= 81,549.7$$

$$= 81,549$$

3. P
V. Imp

(*) How many deposits of Rs 25000 each should Dr. Thakur make each month (or year) so that the final accumulated amount will be 10,00,000 if the bank interest rate is 12%.

Solⁿ

$$Fv = 10,00,000$$

$$i: 12\% = 0.12$$

$$\text{Monthly deposit} = 25,000$$

$$n = ?$$

$$Fv = A \left[\frac{(1+i)^n - 1}{i} \right]$$

$$10,00,000 = 25,000 \left[\frac{(1 + \frac{0.12}{12})^n - 1}{\frac{0.12}{12}} \right]$$

$$40 \times 12 = 1 + \dots$$

$$= 4^{0.4} = (1.01)^n - 1 \text{ on } 1.4 = (1.01)^n$$

$$\text{on } n = \log_{(1.01)} (1.4) \Rightarrow n = \log_{1.01} (1.4)$$

$$= 33.81$$

$$= 33.81$$

$$\Rightarrow 34$$

Use this

Alternatively

$$Fv = A \left[\frac{(1+i)^n - 1}{i} \right]$$

$$i_m = (1+i)^{1/12} - 1$$

$$= (1.12)^{1/12} - 1$$

$$= 0.00948$$

$$10,00,000 = 25,000 \left[\frac{(1.00948)^n - 1}{0.00948} \right]$$

$$\text{on } 40 \times 0.00948 + 1 = (1.00948)^n$$

$$\text{on } 1.3792 = (1.00948)^n$$

$$\text{on } n = \log_{(1.00948)} (1.3792)$$

$$= 34.07$$

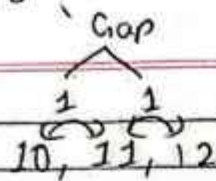
$$\Rightarrow 34$$

same amount for same

19.12.12

Date
Page

(*) Gradient



(*) Arithmetic gradient (Uniform gradient)
Linear
→ PV, FV

Imp $FV = A \left[\frac{(1+i)^n - 1}{i} \right] + \frac{G}{i} \left[\frac{(1+i)^n - 1}{i} \right]$

A = First years deposit

G = incremental amount (every year)

Never
works

$PV = A \left[\frac{(1+i)^n - 1}{i(1+i)^n} \right] + \frac{G}{i^2(1+i)^n} \left[(1+i)^n - 1 - n \right]$

PV (Ques)

* A ~~question~~ person is planning for his ~~retire~~ retired life and ~~hears~~ ^{hears} to ~~g~~ more years to retire. He would like to deposit some portion of his salary, which is Rs 5000 at the end of 1st year and thereafter he wishes to deposit the amount with an annual increase of Rs. for the ^{next} 9 years with an interest of 15%. Find the total amount at the end of 10th year.

Q for the next 9 yrs so, 1 year ~~period~~ ^{period} in

$G = 10$ $A = 5000$, $G = 1000$

$n = 10$

$i = 15\% = 0.15$

$FV = A \left[\frac{(1+i)^n - 1}{i} \right] + \frac{G}{i} \left[\frac{(1+i)^n - 1}{i} \right] - \frac{2G}{i}$

$= 5000 \left[\frac{(1+0.15)^{10} - 1}{0.15} \right] + \frac{1000}{0.15} \left[\frac{(1+0.15)^{10} - 1}{0.15} \right]$

$- \frac{10 \times 1000}{0.15}$

$= 6,11,642.53 - 20,2549.37 + 1,70,210$

Answer

Y	1	2	3	4	5	6	7	8	9	10
C.P	5000	6000	7000	8000	9000	10000	11000	12000	13000	14000

$FV = 5000 \times (1.15)^9 + 6000 \times (1.15)^8 + 7000 \times (1.15)^7 + 8000 \times (1.15)^6 + 9000 \times (1.15)^5 + 10000 \times (1.15)^4 + 11000 \times (1.15)^3 + 12000 \times (1.15)^2 + (1.15)^1 \times 13000 + (1.15)^0 \times 14000$

Ex) Geometric Gradient $r \neq g$

$$P_v = \frac{A}{r-g} \left[1 - \left(\frac{1+g}{1+r} \right)^n \right] \quad \text{if } r \neq g$$

$$P_v = \frac{A \times n}{1+r}$$

A) first cash flow

r = interest rate

g = growth rate

n = no. of year

- (x) A company expects to receive a series of payment starting at Rs 5000 at year 1 growing at 4% per year for 10 years. If the discount rate (i.e. interest rate) is 7%. What is the present value of her

Solⁿ

$$g = 4\% = 0.04$$

$$r = 7\% = 0.07$$

$$A = \text{Rs } 5000$$

$$P_v = ?$$

hr

$$P_v = \frac{5000}{0.07 - 0.04} \left[1 - \left(\frac{1.04}{1.07} \right)^{10} \right]$$

Future value if $r \neq g$

$$F_v = A \times \left[\frac{(1+r)^n - (1+g)^n}{r-g} \right]$$

if $r = g$

$$F_v = A \times n \times (1+r)^{n-1}$$

IMP Most repeated

- * Ramesh, a computer engineer is planning to deposit 20% of his total salary which is 2,50,000 per year, each year in mutual fund. He wants to dep. He expects 7% salary increase each year for 15 yrs. If the mutual fund will give 7% annual return what will the ~~sum~~ ^{sum} amount at the end of 15 yrs. and if salary increases by Rs. 25,000 per year what will the amount amount.

Solⁿ 1st case

First case, salary increasing 7% every year assuming depositing at the end of every year.

Salary

- N/A
- 1) $2,50,000 \times 1.07 = 267,500$ end year salary $\rightarrow$ 7% Deposit 20% $\rightarrow$ 53,500 deposit
 - 2) $267,500 \times 1.07 = 286,225$ Deposit 20% $\rightarrow$ 57,245
 - 3) $286,225 \times 1.07 = 306,260.75$ Deposit 20% $\rightarrow$ 61,252.15
- first year ka deposit nikalni

$$A = 53,500 \quad F_v = A \times \left[\frac{(1+i)^n - (1+g)^n}{i - g} \right]$$

$$g = 7\%$$

$$i = 10\%$$

$$F_v = ? = 53,500 \times \left[\frac{(1+0.1)^{15} - (1+0.07)^{15}}{0.1 - 0.07} \right]$$

$$n = 15 \text{ yrs}$$

$$= 68,48,976$$

$$= 25,29,153$$

2nd case

Salary increases by Rs 25,000 every year assuming depositing at end of every year.

N.A. 1st year salary $(25000 + 25000) = 27500 \rightarrow 55,000$
 2nd year " $27500 + 25000 = 30000 \rightarrow 60,000$
 3rd year " $30000 + 25000 = 32500 \rightarrow 65,000$

Since the amount of deposit increases by 5000 every year this is the case of arithmetic gradient.

$$F_v = A \left[\frac{(1+i)^n - 1}{i} \right] + G \left[\frac{(1+i)^n - 1}{i^2} \right] - \frac{nG}{i}$$

$$A = 5000 \text{ } 55,000$$

$$g = 5000$$

now $\rightarrow$ end $\Rightarrow 1+i$

now $\rightarrow$ now $\Rightarrow$ no change in formula

$$i = 10\% = 0.1$$

$$n = 15 \text{ yrs}$$

$$F_v = 55000 \left[\frac{(1+0.1)^{15} - 1}{0.1} \right] + \frac{5000}{0.1} \left[\frac{(1+0.1)^{15} - 1}{0.1} \right]$$

$$= 15 \times 5000$$

$$= 25,86,110.578$$

(X) What happens if he deposits now (beginning of the year)

First case

- 1) First year 2500000 Deposits 20% $\rightarrow 50,000$
- 2) 2nd year $(250,000 \times 1.07) = 267,500 \rightarrow$ Deposit 20% $\rightarrow 53,500$
- 3) 3rd year $(267,500 \times 1.07) = 286,225 \rightarrow$ Deposit 20% $\rightarrow 57,245$

$$A = 1^{\text{st}} \text{ year deposit} = 50,000$$

$$F_v = 50000 \left[\frac{(1+0.1)^{15} - 1}{0.1} \right] + \frac{5000}{0.1} \left[\frac{(1+0.1)^{15} - 1}{0.1} \right]$$

$$= 15 \times 5000 \times (1+0.1)$$

$$= 26,69,970.985 \text{ } n=7$$

2nd case

$$F_v = \left[A \times \frac{(1+r)^n - (1+g)^n}{r-g} \right] \times (1+i)$$
$$= \left[50,000 \times \frac{(1+0.1)^{15} - (1+0.07)^{15}}{0.1 - 0.07} \right] \times 1.1$$
$$= 26,00,069$$

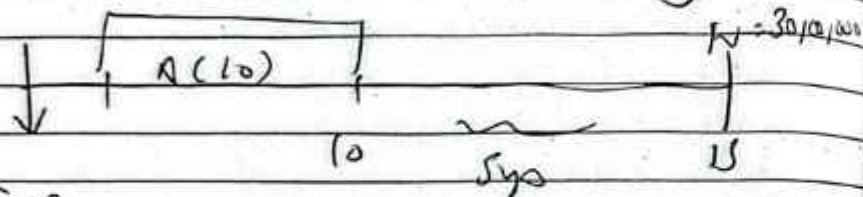
2nd case

- i) First year 2,50,000 Deposits 20% $\Rightarrow$ 50,000
ii) 2nd year 2,75,000 ' ' $\Rightarrow$ 55,000

$$A = 50,000$$

$$F_v = \left[50,000 \left[\frac{(1.1)^{15} - 1}{0.1} \right] + 5000 \left[\frac{(1.1)^{15} - 1}{0.1} \right] \right. \\ \left. - \frac{15,25,000}{0.1} \right] \times 1.1$$
$$= 26,69,972 \text{ Ans}$$

Q Mr. Adhikari wants to have Rs 30,00,000 for the study of his son after the period of 15 yrs. How much rupees does he have to deposit end of each year for 10 continuous year in saving account that earns 10% interest annually



Solⁿ

Given,

$$FV = 30,00,000$$

$$\text{no. of year } (n) = 15 \text{ yrs}$$

$$\text{interest } (i) = 10\% = 0.1$$

every year deposit for 10 yrs = ? 10 yrs = ?

Step 1:

Discounting the 15th year value to 10th year.

$$\text{Value of 10th year} = \frac{30,00,000}{(1+0.1)^5} = 18,62,763$$

Step 2:

Calculation of every year deposit for 10 yrs by using the following formula

⇒ 3 more ques from PV exam - exam

$$FV = A \left[\frac{(1+i)^n - 1}{i} \right]$$

$$18,62,763 = A \left[\frac{(1.1)^{10} - 1}{0.1} \right]$$

$$\begin{aligned} A &= 18,62,763 \times \frac{0.1}{(1.1)^{10} - 1} \\ &= 1,16,879.8 \end{aligned}$$

* Causes of depreciation - 5 marks

- Purchase price $\rightarrow$ zero year, original cost - the cost of assets
- Annual depreciation amount
- Book value $\rightarrow$ value of an asset after deducting annual depreciation
First year $\rightarrow$ purchase price $\rightarrow$ Annual depreciation
- Salvage value $\rightarrow$ value of an asset in n^{th} year
- no. of years (n)
- Depreciation methods.

* Types of depreciation

- N/A
1. Straight line depreciation method
(Fixed installment depreciation method)

$\rightarrow$ It is the simplest

$$\text{Annual depreciation (AD)} = \frac{\text{cost of Asset} - \text{Salvage Value}}{\text{no. of years (useful life)}}$$

Eg

- * A machine costing of Rs. 1,20,000 having estimated salvage value of Rs 20,000

1st year beginning book value = cost of machine

and 10 years useful life. Calculate annual depreciation & amount and book value under st. line depreciation method.

Solⁿ:

Calculation of annual depreciation amount (AD)
we have:

$$AD = \frac{\text{cost of machine} - \text{salvage value}}{\text{Useful life}}$$

$$= \frac{1,20,000 - 20,000}{10}$$

$$= \text{Rs } 10,000$$

Calculation of annual book value:

Year	Beginning Book value	Annual depreciation	Ending book value
1	1,20,000	10,000	1,10,000
2	1,10,000	10,000	1,00,000
3	1,00,000	10,000	90,000
4	90,000	"	80,000
5	80,000	"	70,000
6	70,000	"	60,000
7	60,000	"	50,000
8	50,000	"	40,000
9	40,000	"	30,000
10	30,000	"	20,000

Imp Gov. approved
 2. Diminishing value balance method
Declining
Maths method

- (*) A machine costing Rs 200,000 having estimated salvage value of Rs 30,000 and 6 yrs useful life. Calculate A.D amount and book value under diminishing balance method with 20% depreciation rate.

Solⁿ

Given,

Salvage value = Rs 30,000, n = 6 yrs

Year	Beginning Book value	A.D	E.B.V
1	200,000	40,000	1,60,000
2	1,60,000	32,000	1,28,000
3	1,28,000	25,600	1,02,400
4	1,02,400	20,480	81,920
5	81,920	16,384	65,536
6	65,536	13,107.2	52,428.8

$$R = \left[1 - \sqrt[n]{\frac{s}{\text{Original cost (cost of the asset)}}} \right] \times 100$$

- (*) Dependency is Diminishing balance method where rate is not given

Solⁿ:-

Example:

- A machine costing of Rs 36,000, at estimated salvage value at end of 5th year is Rs 6,000. Calculate annual depreciation amount & book value under diminishing balance method.

Ans

Original cost = Rs 36,000

n = 5 yrs

Salvage value (s) = 6000

We have,

$$\text{rate (R)} = \left[1 - \sqrt[5]{\frac{6000}{36000}} \right] \times 100$$

$$= 30.12\%$$

Calculation of depreciation amount and book value, D.F.P @ 30.12%.

Year	Beginning Book value	A.D	E.B.V
1	36,000	10843.2	25156.8
2	25156.8	7577.22	17579.58
3	17579.58	5297.96	12281.62
4	12281.62	3699.95	8581.67
5	8581.67	2585.52	5996.15

V.V.
Imp

If rate is missing:-
Since interest rate is not given
assume :: 10%.

(*) Sinking fund method (last year value must equal to salvage value)

Step 1: Calculation of fixed depreciation amount with the help of the following formula.

$$F.D.A = \frac{I-S}{(F_{v,i}) i \% n y}$$

Most repeated

* Compute the depreciation charge (annual) and book value of each year by using sinking fund method with the following information.

Initial cost of asset = Rs 1,00,000 (I)

Salvage value (S) = 20,000

Life of Asset (n) = 8 yrs

Interest rate (i) = 12%

Soln

Step 1:

Calculation of fixed depreciation amount:

$$F.D = \frac{I-S}{F_{v,i} 12\% \cdot 8 y}$$

$$= \frac{1,00,000 - 20,000}{12.2897}$$

$$= 6504.22$$

Net depreciation = $FD \times (1+i)^{k-1}$
where, k = particular year
1st year = 1, 2nd year = 2

Step 2: Calculation of every year depreciation amount and book value:

Year	Reg. Book Value	Fixed Depreciation	Net Depreciation	Ending value
1	1,00,000	6504.22	6504.22	93495.78
2	93495.78	6504.22	7289.72	86211.06
3	86211.06	6504.22	8158.69	78052.37
4	78052.37	"	9137.96	68914.41
5	68914.41	"	10234.52	58679.89
6	58679.89	"	11462.66	47217.23
7	47217.23	"	12838.18	34379.05
8	34379.05	"	14378.76	20000.29

$$1 \rightarrow 6504.22 \times (1+0.12)^0$$

$$2 \rightarrow 6504.22 \times (1+0.12)^1$$

Nm
calculator

$$FD \times (1+i)^{k-1}$$

fixed
one

Not so imp

$$(*) d_k = F.D \times F_{v,i} i \% (k-1) y$$

$$= \left[\frac{I-S}{F_{v,i} i \% n y} \right] \times F_{v,i} i \% (k-1) y$$

7th year

$$= 6504.22 \times F_{v,i} 12\% \cdot 6 y$$

$$= 6504.22 \times 12.2897 \cdot 6 y$$

Qalib

$I \rightarrow S \Rightarrow I$ (1000000)
Remaining year has 10 but you have 10 full year.

Date _____
Page _____

(*) Calculation of ending book value for the particular year ~~at the end of year~~

$$BV_k = I - [F.D \times (F_v \text{ i.e. } k \text{ yrs})]$$

$$= 10,00,000 - [6504.22 \times 10.0890]$$

$$= 34,378.92$$

V.V. Imp Sure for U.T.

Sum of year digit method (SOYD)

- A machine costing of Rs. 1,50,000 having estimated salvage value of Rs. 15,000 and 10 yrs useful life. Calculate annual depreciation amount and net value under SOYD method

Solⁿ

Given,

$$I = \text{Rs. } 1,50,000$$

$$S = 15,000$$

$$n = 10 \text{ yrs}$$

1 Calculation of SOYD

$$\text{Sum of } n \text{ yrs} = \frac{n(n+1)}{2}$$

$$\text{Annual depreciation} = (I - S) \times \frac{\text{rem. yrs}}{\text{SOYD}}$$

Date _____
Page _____

$$\text{SOYD} = \frac{n(n+1)}{2} = \frac{10 \times 11}{2} = 55$$

Step 1: Calculation of annual depreciation and book value:

Year	Beg. Book value	Annual dep. $(I - S) \times \frac{\text{rem. yrs}}{\text{SOYD}}$	Ending book value
1	1,50,000	$(1,50,000 - 15,000) \times \frac{10}{55} = 18,181.82$	131,818.19
2	131,818.19	$13,363.55$	118,454.64
3	118,454.64	14,545.45	103,909.19
4		12,727.27	91,181.92
5		10,909.09	80,272.83
6		9,090.91	71,181.92
7		7,272.73	63,909.19
8		5,454.55	58,454.64
9		3,636.36	54,818.28
10		1,818.18	53,000.10

5) double rate declining method $\rightarrow$ double line declining method:

Step 1: Calc. of depreciation rate (rate of st. line)

$$\text{rate} = \left(\frac{1}{n} \times 100 \right) \times 2$$

$$\text{If } n \text{ is } 10 \text{ yrs, } r = \frac{1}{10} \times 100 \times 2 = 20\%$$

Step 2: Use diminishing balance method with same above rate and calculate depreciation amount & book value

Ex: Consider the following accounting for computer system

initial cost of the computer = Rs 20,000

Useful life = 5 yrs

Compute the annual depreciation amount and resulting book values by using double rate declining method.

Step 1: Calculation of rate

$$r = \left(\frac{1}{n} \right) \left(\frac{1}{n} \times 100 \right) \times 2 = \left(\frac{1}{5} \times 100 \right) \times 2 = 40\%$$

Dep 2:

Yr.	Beginning book value	Depreciation at rate 40%	Ending book value
1	20,000	4000	16000
2	16000	1840	14160
3	14160	1824	12336
4	12336	2384	9952
5	9952		7216

Imp

6) St-line + Avg. interest method / switch over method.

Ex: Consider the following information for a machine

initial cost = Rs 4000

useful life = 10 yrs

salvage value (S) = 0

depreciation rate = 20%

Calc. the annual depreciation amount & book value by using st-line + avg interest method.

Calc. of depreciation amount & book value

Yr	Beg. book value	Dep. SL method	Dep. DB method 20%	Dep. reduction	Ending book value
1	4000	400	800	800	3200
2	3200	355.55	640	640	2560
3	2560	320	512	512	2048
4	2048	292.57	409.6	409.6	1638.4
5	1638.4	273.06	327.6	327.6	1310.8
6	1310.8	262.16	262.16	262.16	1048.6
7	1048.6	262.16	262.16	262.16	786.4
8	786.4	"	209.72	"	576.68
9	576.68	"	157.29	"	419.39
10	419.39	"	125.87	"	293.52
11	293.52	"	73.42	"	220.10

7 Service output method / unit of production method.

Ex A machine costing of Rs 50,000 is estimated to have Rs 10,000 salvage value when replaced after 30,000 hours of use. Find its depreciation rate per hour its use and depreciation amount and book value after 10,000 hrs of operation.

Solⁿ:

$$\text{Calc. of rate} = \frac{\text{cost} - \text{salvage}}{\text{life}} = \frac{50,000 - 10,000}{30,000} = 1.33$$

Now, depreciation amount after use of 10,000 hrs.

$$10,000 \times 1.33 = 13,330$$

$$\text{Book value} = \text{cost} - \text{depreciation} = 50,000 - 13,330 = 36,670$$

Cost = Rs 50,000	rate = $\frac{\text{cost} - \text{sv}}{\text{no. of hrs}} = \frac{50,000 - 10,000}{30,000} = 1.33$
S.V = Rs 10,000	
Total hrs = 30,000	
Used hrs = 10,000	Dep amount after 10,000 hrs = $10,000 \times 1.33 = 13,330$
Dep rate = 1	
Book value = ?	Book value = cost - Dep = 36,670

Depreciation

- Causes of depreciation

Imp 2-3 barsha me ek baar

8) The modified accelerated cost recovery system (MACRS):-

Depreciation $\rightarrow 100\%$

If Sys $\rightarrow$

Then,
1st yr $\rightarrow 0.5$ (half)
2nd yr
3rd yr
4th yr
5th yr
6th yr $\rightarrow 0.5$ (half)
Full

• Step-1:

Calc rate

$$\text{DB} \quad \text{SL} \quad \left(\frac{1}{n} \times 100 \right) \times 2$$

If yrs

3
5
7
10
SL = $\frac{1}{n} \times 100$ DB $\times 2$
St. line Diminishing

else if 15 yrs

$$\text{DB} = \text{SL} \times 1.5$$

4 choti
Imp

If bgyo vane table harmi
last page MACRS Table bata

Date _____
Page _____

Eg: A tax payer wants to place in service of
Rs 10,000 asset that is assigned to be
5 yrs class. Compute the every year MACRS
percentage, depreciation amount and book
value for the asset.

we have
to go
bgy

Solⁿ

Calculation of MACRS %

~~SL rate~~

$$SL \text{ rate} = \frac{1}{n} \times 100$$

$$= \frac{1}{5} \times 100\%$$

$$= 20\%$$

$$DB \text{ rate} = 20\% \times 2$$

$$= 40\%$$

Year	MACRS % rate	Depreciation rate selection	Book value
1 (2)	SL $\rightarrow \frac{1}{5} \times 100 \times 0.5 = 10\%$ DB $\rightarrow \frac{1}{5} \times 100 \times 2 \times 0.5 = 20\%$	20%	80%
2	SL $\rightarrow \frac{2}{5} \times 80\% = 17.78\%$ DB $\rightarrow \frac{2}{5} \times 80\% \times 2 = 35.56\%$ (40% of 80)	32%	48%
3	SL $\rightarrow \frac{3}{5} \times 48\% = 13.71\%$ DB $\rightarrow 40\% \text{ of } 48 = 19.2\%$	19.2%	28.8%
4	SL $\rightarrow \frac{4}{5} \times 28.8\% = 11.52\%$ DB $\rightarrow 40\% \text{ of } 28.8 = 11.52\%$	11.52%	17.28%
5	SL $\rightarrow \frac{1}{5} \times 17.28\% = 11.52\%$ DB $\rightarrow 40\% \text{ of } 17.28 = 6.912\%$	11.52%	5.76%
6	SL $\rightarrow \frac{1}{5} \times 5.76\% \times 0.5 = 0.576\%$ DB $\rightarrow 40\% \text{ of } 5.76 = 2.304\%$	5.76%	0

Second part

Calc. of annual depreciation and book
value :-

Year	Beg book value	MACRS	Dep amount	Ending book value
1	10,000	20%	2000	8000
2	8000	32%	2560	4800
3	4800	19.2%	921.6	2880
4	2880	11.52%	331.776	1728
5	1728	11.52%	197.856	576
6	576	5.76%	33.1776	0

Always depreciate the initial cost
10,000

(X) A zinc company bought ^{on} and sold ^{ore} ore buying
land for Rs 20,00,000. The recoverable
resources in the mine were estimated to be
50,00,000 tonnes.

a) If 50,000 tonnes of ore were mined & sold during
the year, what was the depletion charge and
ending book value for 2009.

b) Calculation of cost per ton

$$\text{Per ton cost} = \frac{\text{Total cost}}{\text{No. of tons}} = \frac{20,00,000}{50,00,000}$$

Calculation of depletion amount for 5000 tonnes
 $= 50000 \times 7 = \text{Rs } 20,00,000$

Ending book value at end of 2009
 $= 20,100,000 - 20,00,000$
 $= \text{Rs } 18,00,000$

b) Suppose at the end of 2009, the results were revalued and found to be only 4,00,000 tonnes. If 50,000 tonnes of ores were mined & sold in during 2010 what is depreciation charge for 2010 and book value.

Soln

Calc. of cost per tonne after revaluation:

$$= \frac{18,00,000 \times 50,000}{4,00,000} = 2,25,000$$

• Basic method of Engineering economic studies

1 Equivalent worth method

└ Present worth method (Pw)
 └ Future " " (Fw)
 └ Annual " " (Aw)

2 Rate of return methods → (i) → 1 question
 (a) IRR method
 (b) ERR "

(3) Payback period → (ii) → 1 question
 a) Simple
 b) Discounted

(4) Benefit cost ratio method → (iii) → 1 question
 a) Conventional BC ratio
 b) Modified BC ratio

(X) Parameter of project

- 1 Initial investment (P_i) (one time)
- 2 Annual Revenue (AR) (Annuity) (every year)
- 3 Annual cost (AC) (Annuity) (every year)
- 4 Net annual amount revenue (NAR) (Annuity)
 every year $\Rightarrow AR - AC$
- 5 Salvage value F_v at the end of project life
- 6 MARR interest rate (Minimum Attractive Rate of Return)

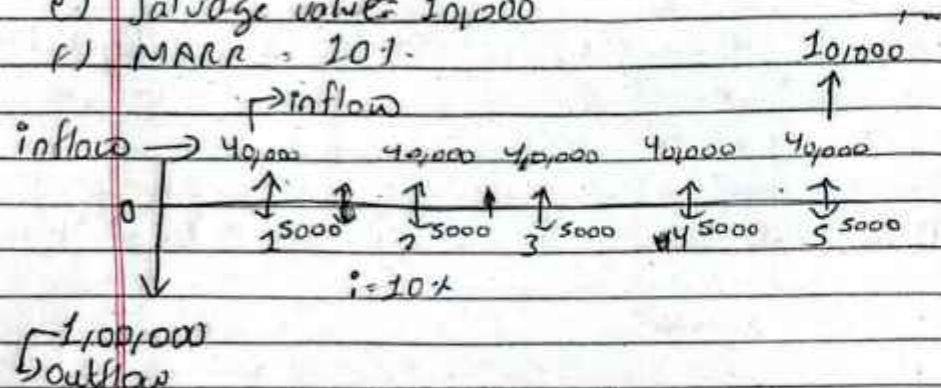
CFD → 1 marks

Date _____
Page _____

Compulsory

(*) CFD (Cash Flow Diagram)

- Initial investment = 1,00,000
- Annual revenue = 40,000
- Annual cost = 5,000
- Useful life = 5 yrs
- Salvage value = 10,000
- MARR = 10%



Net annual revenues = 40,000 - 5,000
= 35,000

(i) Equivalent worth method

- PW
- FW
- AW

if PW is positive → Accept the project
elif PW is negative → Reject the project

(i) Present worth method (PW method)

$$PW_{i\%} = -$$

$$PW_{i\%} = -II + NAR \times \frac{pvifa_{i\%} \cdot n}{(1+i)^n} + \frac{Sw}{(1+i)^n}$$

$$PW_{10\%} = -1,00,000 + 35,000 \times \frac{pvifa_{10\%} \cdot 5}{(1+0.1)^5} + \frac{10,000}{(1+0.1)^5}$$

$$= -1,00,000 + 35,000 \times 3.7901 + \frac{10,000}{(1+0.1)^5}$$

$$= 38,887.21$$

Since, the PW of the project is positive i.e. 38,887.21, the project should be accepted.

(*)

Cashflow

Cash inflow

- Revenue
 - Annual revenue
 - Salvage value

Cash outflow

- Investment and cost
 - initial investment
 - Annual cost

(*) Pw method for uneven cash flow :

- Calculate Pw of the following project :

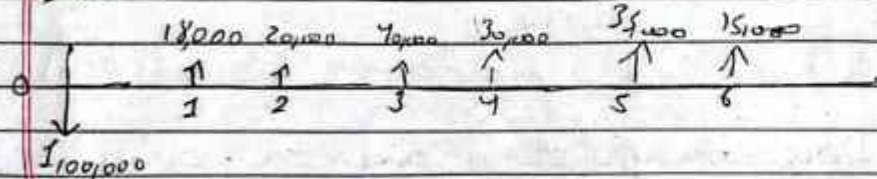
initial investment = 1,00,000

Cash flow

Year	1	2	3	4	5	6
Cash flow	18,000	29,000	40,000	30,000	35,000	15,000

interest value (MARR) = 10%

CFD



$$Pw_{10\%} = -1,00,000 + \frac{18,000}{(1.1)^1} + \frac{20,000}{(1.1)^2} + \frac{40,000}{(1.1)^3} + \frac{30,000}{(1.1)^4} + \frac{35,000}{(1.1)^5} + \frac{15,000}{(1.1)^6}$$

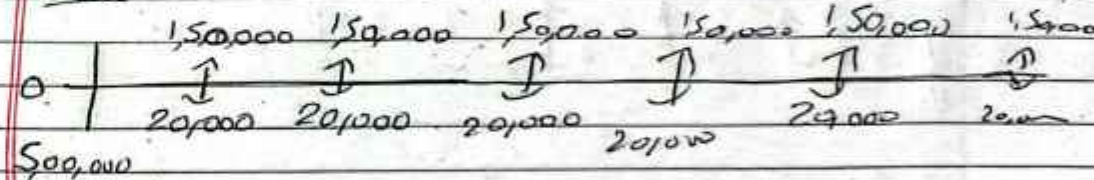
$$= 15,808.13 / 12.624$$

Fw

(*) Evaluate the following project under Fw formulation.

- Initial investment = 5,00,000
- Useful life = 6 yrs
- Annual revenue = 1,50,000
- Annual cost = 20,000
- MARR = 8%
- Salvage value = 50,000

CFD



$$NAR = 1,50,000 - 20,000 = 1,30,000$$

$$Fw_{8\%} = -5,00,000 + 1,30,000 \times Fw_{8\%} + 50,000$$

$$= -5,00,000 \times (1.08)^{-6} + 1,30,000 \times Fw_{8\%} + 50,000$$

$$= -5,00,000 \times (1.08)^{-6} + 1,30,000 \times 7.3351 + 50,000$$

$$= 2,10,229.8385$$

Since, calculated Fw is positive : 2,10,229.8385 project should be accepted.

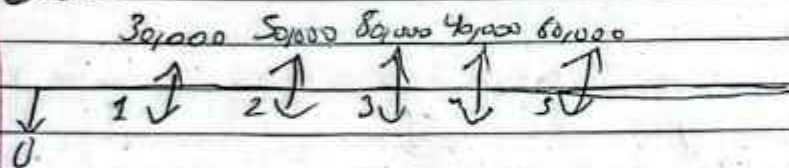
* Fw for uneven cashflow

- Evaluate following project under Fw formulation

Year	0	1	2	3	4	5
Cashflow	2,00,000	30,000	50,000	80,000	40,000	60,000

$$MARR = 10\%$$

② IFD



$$\begin{aligned}
 Fw_{10\%} &= -2,00,000 \times (1.1)^0 + 30,000 \times (1.1)^{-1} \\
 &\quad + 50,000 \times (1.1)^{-2} + 80,000 \times (1.1)^{-3} \\
 &\quad + 40,000 \times (1.1)^{-4} + 60,000 \times (1.1)^{-5} \\
 &= -10,829
 \end{aligned}$$

Since the calculate Fw of the project is '-ve' i.e. -10,829, this project should be rejected

① AW method:

$$AW\% = - \frac{II}{PVI(A, i, n)_{yrs}} + NAR + \frac{SV}{FVI(A, i, n)_{yrs}}$$

Eg: Evaluate the following project under AW formulation.

$$II = 10,00,000$$

$$\text{Annual revenue} = 1,50,000$$

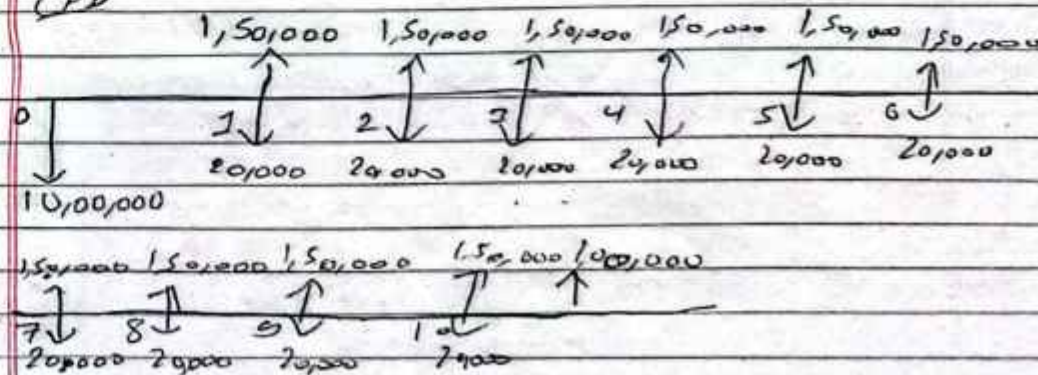
$$\text{Annual cost} = 20,000$$

$$\text{Salvage value} = 1,00,000$$

$$\text{Useful life} = 10 \text{ yrs}$$

$$MARR = 12\%$$

② CF



$$\text{Ans 12: } = - \frac{10,00,000}{\text{PVIFA } 12\% \cdot 10\text{yr}} + \frac{1,30,000 + \frac{1,00,000}{1.12^{10}}}{\text{PVIFA } 12\% \cdot 10\text{yr}}$$

$$= - \frac{10,00,000}{5.6502} + \frac{1,30,000 + \frac{1,00,000}{17.5487}}{5.6502}$$

$$= -41,286.45$$

So, project should be rejected.

Sum

2. Rate of return method

- a) Internal rate of return (IRR) is rate where ^{V.Imp}
It is concerned with PW method. $PW = 0$

$$PW = -I_0 + \text{NAR} \times \text{PVIFA } i, n + \frac{S.V.}{(1+i)^n}$$

$$\text{Initial investment} = 1,00,000$$

$$\text{Annual revenue} = 1,40,000$$

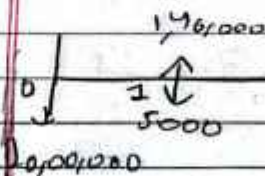
$$\text{Annual cost} = 5,000$$

$$\text{Salvage value} = 5,000$$

$$\text{Useful life} = 10 \text{ yr}$$

$$\text{MARR} = 10\%$$

CFD



$$0 = -10,00,000 + 1,35,000 \times \frac{(1+i)^{10} - 1}{i(1+i)^{10}} + \frac{5,000}{(1+i)^{10}}$$

Hints

$$\text{Factor} = \frac{\text{Initial investment}}{\text{Avg net annual revenue}}$$

$$\text{Avg net annual revenue}$$

$$= \frac{10,00,000}{1,35,000 + \frac{50,000}{10}}$$

7.14

for

$n = 10$ yrs to 7.14 to reach ka na karna
for 10, then $i = 7\%$.

Try at 7%.

$$Pw 7\% = -10,00,000 + 1,35,000 \times P_{vifA} 7\% 10 \text{ yrs} \\ + \frac{50,000}{(1.07)^{10}}$$

$$= -10,00,000 + 1,35,000 \times 7.0236 +$$

$$= -26,396.5353$$

Try at 6%.

7.3601

$$Pw 6\% = -10,00,000 + 1,35,000 \times P_{vifA} 6\% 10 \text{ yrs} \\ + \frac{50,000}{(1.06)^{10}}$$

$$= 21,533.73885$$

Interpolating for IRR

$$IRR = LR + \left[\frac{Pw \text{ of } LR}{Pw \text{ of } LR - Pw \text{ of } HR} \times Diff \right]$$

$$= 6\% + \left[\frac{253 + 21531.49}{21531.49 + 26396.53} \times 1 \right]$$

$$= 6.49\%$$

IRR for uneven cash flow

Evaluate following under IRR function.

Year	0	1	2	3	4	5	6
Cashflow	5,00,000	120,000	140,000	100,000	90,000	30,000	1,00,000

MARR = 12%

for
(FID)

	120,000	140,000	100,000	90,000	30,000	1,00,000
	↑	↑	↑	↑	↑	↑
	1	2	3	4	5	6
0						

Minb = JJ

Average cash flow

$$= \frac{500,000 + \text{Cashflow sum from 1st year to 6}}{6} \\ = 4.74$$

which equals to 10% in PwLA.

Date _____
Page _____

$$P_w 10\% = -500,000 + \frac{1,200,000}{1.1} + \frac{1,400,000}{(1.1)^2} + \frac{1,000,000}{(1.1)^3} + \frac{900,000}{(1.1)^4} + \frac{1,300,000}{(1.1)^5} + \frac{1,100,000}{(1.1)^6}$$

$$= \text{4207.98}$$

$$P_w 12\% = -500,000 + \frac{1,200,000}{1.12} + \frac{1,400,000}{(1.12)^2} + \frac{1,000,000}{(1.12)^3} + \frac{900,000}{(1.12)^4} + \frac{1,300,000}{(1.12)^5} + \frac{1,100,000}{(1.12)^6}$$

$$= -23,350.43$$

Interpolating for IRR,

$$IRR = LR + \left[\frac{P_w \text{ of } LR}{P_w \text{ of } LA - P_w \text{ of } LR} \right] \times D, 1\%$$

$$= 10\% + \left[\frac{4207.98}{4207.98 + 23350.43} \right] \times 2\%$$

Date _____
Page _____

$$= 10\% + 0.305$$

$$= 10.305\%$$

Decision

if IRR is greater than MARR accept the project.

if IRR is less than MARR reject the project.

Since, calculated IRR of the project is less than given MARR, the project is rejected.

investment $\rightarrow 0$ years $\rightarrow$ This must be dragged to year 1
 One month $\rightarrow 1$ year to subtract it with one line last year return.

④ Calculate IRR and also show unrecovered investment balance in table and graph.
 (4+2+2)

initial investment = 1,00,000

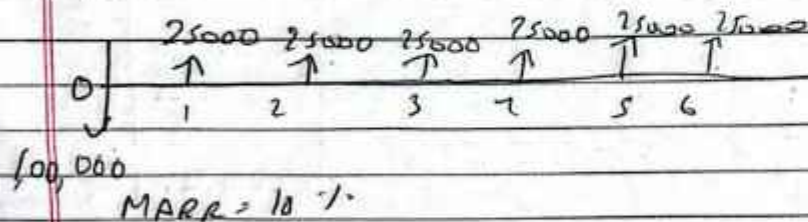
Annual income = 25,000

Annual Salvage value = 5,000

MARR = 10%

Useful life = 6 yrs

for
CFD



$$H_{10\%} = \frac{1,00,000}{25,000 + \frac{5,000}{6}}$$

$$= 3.87$$

we look at table and found at 14%.

Present worth rule

$$Pw_{14\%} = -1,00,000 + 25,000 \times Pw_{14\%} \text{ for } 6 \text{ yrs} + \frac{5,000}{(1.14)^6}$$

$$= -1,00,000 + 25,000 \times 3.8887 + \frac{5,000}{(1.14)^6}$$

$$= -1,00,000 + 97,217.5 - 504.56$$

Try at 13%

$$Pw_{13\%} = -1,00,000 + 25,000 \times Pw_{13\%} \text{ for } 6 \text{ yrs} + \frac{5,000}{(1.13)^6}$$

$$= -1,00,000 + 25,000 \times 3.9972 + \frac{5,000}{(1.13)^6}$$

$$= -1,00,000 + 99,930 + 504.56$$

Interpolation for IRR

$$IRR = L + \frac{Pw_{13\%} - Pw_{14\%}}{Pw_{13\%} - Pw_{14\%}} \times (14\% - 13\%)$$

$$= 13\% + \frac{2,341.59}{2,341.59 + 504.56} \times 1\%$$

$$= 13.82\%$$

$$IRR = 13.82\%$$

Date _____
Page _____

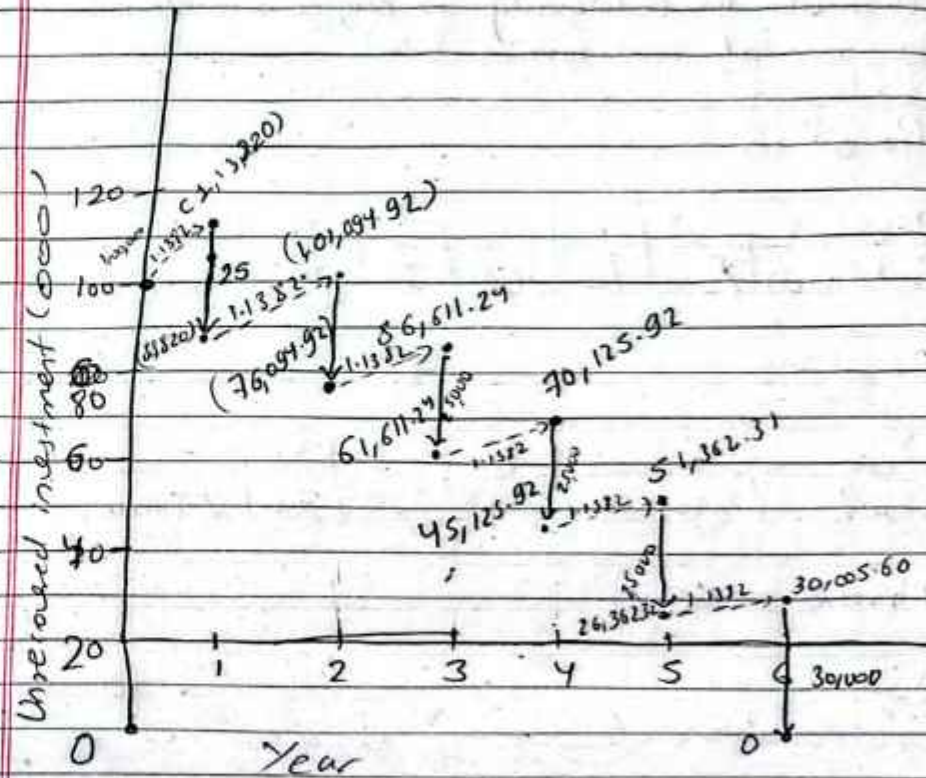
(*) Tabular form

* Uncovered investment calculation in tabular form:

Year	Cashflow	Uncovered investment at the beginning	Value at end after considering IRR	Uncovered investment at the end
0	-1,00,000	-	-	-1,00,000
1	25,000	-1,00,000	$-1,00,000 \times 1.1382 = -1,13,820$	-88,820
2	25,000	-88,820	$-1,13,820 \times 1.1382 = -1,30,094.92$	-76,094.92
3	25,000	-76,094.92	$-1,30,094.92 \times 1.1382 = -1,48,611.29$	-61,611.29
4	25,000	-61,611.29	$-1,48,611.29 \times 1.1382 = -1,69,125.92$	-45,125.92
5	25,000	-45,125.92	$-1,69,125.92 \times 1.1382 = -1,92,362.31$	-26,362.31
6	30,000	-26,362.31	$-1,92,362.31 \times 1.1382 = -2,18,005.60$	-5.60

Uncovered investment

(*) Uncovered investment before graph



(*) Incremental IRR

Example:-

Evaluate the following 2 projects under incremental assumption IRR.

Project A

Year	0	1	2	3	4	5
Cashflow	1000	100	120	130	140	1200

Project B

Year	0	1	2	3	4	5
Cashflow	2000	180	210	250	300	2040 2400

MARR = 10%.

Solⁿ

Calculation of incremental cashflow

Year	0	1	2	3	4	5
Cashflow	2000-1000=1000	180-100=80	210-120=90	250-130=120	300-140=160	2400-1200=1200

Calc. of IRR for incremental cashflows

MARR 11 → Charai invest Value 11%
MARR 11 → Charai " "

Try at 12%.

$$B_w 12\% = -1000 + \frac{80}{1.12} + \frac{90}{(1.12)^2} + \frac{120}{(1.12)^3} + \frac{160}{(1.12)^4} + \frac{1200}{(1.12)^5}$$

$$= 11.184$$

Try at 13%.

$$P_w 13\% = -1000 + \frac{80}{(1.13)} + \frac{90}{(1.13)^2} + \frac{120}{(1.13)^3} + \frac{160}{(1.13)^4} + \frac{1200}{(1.13)^5}$$

$$= -26.11$$

Interpolating for IRR

$$IRR = LR + \frac{P_w \text{ of Lower Rate}}{P_w \text{ of LR} - P_w \text{ of Higher rate}} \times \text{diff.}$$

$$= 12 + \frac{11.184}{11.184 + 26.11} \times 1$$

12.29%
B should be selected as IRR > MARR



- Select project with highest investment if incremental IRR is greater than given MARR and vice versa.

⊗ External rate of return (Modified internal rate of return)

- Following steps are used to calculate ERR:

Step 1: Discounting all cash outflows into zero year.

Step 2: Compounding all cash inflows into final year.

Step 3: ERR is calculated from the following way

Consider the following cashflow of the project.

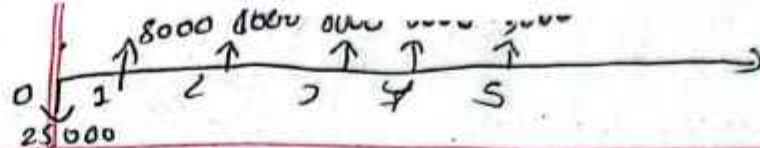
$$I = 25,000$$

$$NAR = 8000$$

$$S.V. = 5000$$

$$\text{Useful life} = 5 \text{ yr}$$

$$MARR = 20\%$$



Step 1

Discounting all cashflow into zero year.
 $= 25,000$

Step 2:

Compounding all cash inflow into 5th yr.
 $= 8000 \times F_{v,i} 20\% 5 \text{ yrs} + 15000$
 $= 64,532.8$

Step 3: Establishing equivalence between step 1 and step 2.

Step 1: $(1 + ERR)^5 \cdot \text{Step 2}$

$$25000 (1 + ERR)^5 = 64,532.8$$

$$(1 + ERR)^5 = \frac{64,532.8}{25,000}$$

$$(1 + ERR)^5 = 2.5813$$

$$ERR = 0.2088$$

$$= 20.88\%$$

ERR and uneven cashflow

(*) Evaluate the following project under ERR formulation.

Year	0	1	2	3	4	5	6
Cashflow	-1,00,000	20,000	30,000	25,000	40,000	15,000	45,000

MARR = 12%.

Solⁿ
CFR

	20000	30k	25k	40k	15k	45k
0	↑	↑	↑	↑	↑	↑
↓	1	2	3	4	5	6
	100,000					

Step 1

Discounting all cash outflows into zero year
= 1,00,000

Step 2

Compounding all cash inflows into the sixth year

$$20,000 \times (1.12)^5 + 30,000 \times (1.12)^4 + 25,000 \times (1.12)^3 + 40,000 \times (1.12)^2 + 15,000 \times (1.12) + 45,000 = 229,551.61$$

Step 3

$$1,00,000 \times (1 + \text{ERR})^6 = 229,551.61$$

or ERR =

$$\text{ERR} = \sqrt[6]{\frac{229,551.61}{1,00,000}} - 1$$

PU

(*)

Year	0	1	2	3	4	5
C.F	-500	202	-x	196	350	451

MARR = 10%, and ERR = 14.1%
Find the value of x

* Payback period (PI)

• Simple payback period

- It does not follow the concept of T.V.M.
(Time value of money)

$$\text{Simple payback period} = \frac{\text{Initial Investment (II)}}{\text{Net annual Revenue (NA)}}$$

Eg

$$II = 1,00,000$$

$$A.R = 25,000$$

$$A.C = 10,000$$

$$S.V = 5,000$$

$$\text{Life(n)} = 10 \text{ yrs } \left. \vphantom{\begin{matrix} II \\ A.R \\ A.C \\ S.V \end{matrix}} \right\} \text{NA}$$

$$\text{MARR} = 10\%$$

$$\text{Simple payback period} = \frac{1,00,000}{25,000 - 10,000} = 4 \text{ yrs}$$

Eg

for uneven cash flow (SPP)

Year	0	1	2	3	4	5	6
C.F	-1,00,000	20,000	30,000	25,000	40,000	15,000	45,000

$$(P.P.) \leftarrow [t]$$

Date _____
Page _____

Calc. of payback period.

Year	cashflow	cumulative cashflow
0	-1,00,000	-1,00,000
1	20,000	-80,000
2	30,000	-50,000
3	25,000	-25,000
4	40,000	+15,000
5	15,000	+30,000
6	45,000	+75,000

payback period [3] years
biching

ignore negative

n.a.

payback period = lower year + cumulative c.f.

$$= \text{lower year} + \frac{\text{cumulative cashflow of lower year}}{\text{cashflow of higher year}}$$

$$= 3 \text{ year} + \frac{25000}{40000}$$

$$= 3.625 \text{ year}$$

T.V.M consider genera

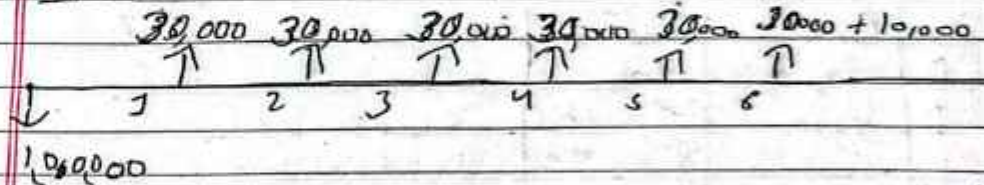
Date _____
Page _____

Sure

(*) Discounted payback period
→ It follows concept of T.V.M

initial investment	1,00,000
Annual renewal	30,000
Annual cost	5,000
Useful life	6 yrs
Salvage value	10,000
MARR	= 10%

CFD



Year	cashflows		Cumulative discounted cashflow
0	-1,00,000		-1,00,000
1	30,000	$\frac{30,000}{(1.1)^1} = 27,272.7$	-72,727.3
2	30,000	$\frac{30,000}{(1.1)^2} = 24,793.38$	-47,934
3	30,000	$\frac{30,000}{(1.1)^3} = 22,539.47$	-25,395.5
4	30,000	$\frac{30,000}{(1.1)^4} = 20,490.40$	-4905.28
5	30,000	$\frac{30,000}{(1.1)^5} = 18,627.61$	+13,723.15
6	40,000	$\frac{40,000}{(1.1)^6} = 22,578.95$	+36,302

Interpolating for payback period

payback period = lower year + $\frac{\text{cumulative CF at lower year}}{\text{Cashflow of H.Y.}}$

$$= 4 \text{ year} + \frac{4905.28}{18,627.67}$$

$$= 4.26 \text{ yrs}$$

④ Uneven cashflow and discounted payback period

Year	Cashflow	Discounted cash flow	Cumulative discounted C.F.
0	-100,000	-100,000	-100,000
1	40,000	$\frac{40,000}{1.1} = 36,364$	-63,636.37
2	60,000	$\frac{60,000}{(1.1)^2} = 49,061$	-14,049.67
3	30,000	$\frac{30,000}{(1.1)^3} = 21,539.77$	+8,489.84
4	75,000	$\frac{75,000}{(1.1)^4} = 50,000$	+58,489.84

$$MARR = 10\%$$

10%

④ Evaluate the following project under IRR formulation.

Year	0	1	2	3	4	5	6	7	8
C.F.	-10,00,000	2,00,000	-10,000	1,50,000	1,30,000	1,60,000	90,000	210,000	240,000

$$MARR = 8\%$$

Hint:

I bato 8 samma jodeno divide by 8

$$i.e. \frac{12,20,000}{8}$$

$$Minb = \frac{10,00,000}{12,20,000} = 6.55$$

Avila table herda (8th yr, 7.14 (kara) ka no)

$$P_{0.5\%} = -10,00,000 + \frac{2,00,000}{1.05} + \frac{-10,000}{(1.05)^2} + \frac{1,50,000}{(1.05)^3} + \frac{1,30,000}{(1.05)^4} + \frac{1,60,000}{(1.05)^5} + \frac{90,000}{(1.05)^6} + \frac{210,000}{(1.05)^7} + \frac{240,000}{(1.05)^8}$$

$$= -36,723.92$$

So, I ch reduce at 3% (high fluctuation value)

$$P_{0.3\%} = -10,00,000 + \frac{2,00,000}{(1.03)} + \frac{-10,000}{(1.03)^2} + \frac{1,50,000}{(1.03)^3} + \frac{1,30,000}{(1.03)^4} + \frac{1,60,000}{(1.03)^5} + \frac{90,000}{(1.03)^6} + \frac{210,000}{(1.03)^7} + \frac{240,000}{(1.03)^8}$$

$$= 55,546.13$$

Interpolation

$$IRR = LR + \frac{P_{0.5\%}}{P_{0.3\%} - P_{0.5\%}} \times \text{diff.}$$

$$= 3 + \frac{55,546.13}{55,546.13 + 36,723.92} \times 2$$

4.20.1.

Size question

(4) BC ratio (Benefit-cost) = $\frac{\text{all benefit}}{\text{all cost}}$

1) Conventional BC ratio we have: cash flows

2) Modified " " i) Initial invest

ii) Annual rev

iii) Annual cost

iv) Salvage value

↓ ques me AW, FW,

PW jo bhanna

subh kismein laas

Concept:

(i) PW formulation

i) Initial investment (PW maina) → cost

ii) Annual revenue (AW) → each year le benefit

business

→ PW la, wala laas → cost

as formula divide

→ go bhanna

(ii) Annual cost (O&M)

→ go par: AW bhanna

iii) Salvage value (FW) → cost (cost
O&M: cost

(1) BC ratio = $\frac{PW(B)}{PW(I) + PW(O\&M) - PW(SV)}$

$$= \frac{PW(B)}{PW(I) + PW(C) - PW(SV)}$$

(2) Modified BC ratio

$$BC = \frac{PW(B) + PW(SV)}{PW(I) + PW(O\&M)}$$

} khas actual go bhanna
pari

$$= \frac{PW(B) - PW(O\&M)}{PW(I) - PW(SV)}$$

} go modified

$$= \frac{PW(B) - PW(C)}{PW(I) - PW(SV)}$$

Example

Formulation

Calculate both type of BC ratio under PW/FW/AW

Evaluate it:

$$II = 2,00,000$$

$$AR = 40,000$$

$$AC = 10,000$$

$$SV = 35,000$$

$$MARR = 10\%$$

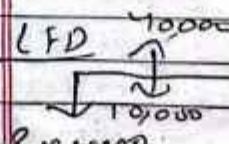
Conventional BC ratio under

PW formulation

we have

$$BC \text{ ratio} = \frac{PW(B)}{PW(I) + PW(C) - PW(SV)}$$

$$= \frac{PW(B) - PW(O\&M)}{PW(I) - PW(SV)}$$



Calculation of $PW(R)$: $40000 \times P_{u,fa} 10\%, 10$
 $= 245784$

$PW(TI) = 200,000$
 $PW(C) = 10,000 \times P_{u,fa} 10\%, 10$
 $= 10,000 \times 6.1446$
 $= 61,446$

$PW(SV) = SV = n^{th} \text{ year } 100$
 $\frac{SV}{(1+i)^n}$
 $= \frac{35000}{(1.1)^{10}} = 13,794.01$

$\therefore BC \text{ ratio} = \frac{P_{u,fa}(R)}{P_{u,fa}(TI) + P_{u,fa}(C) - P_{u,fa}(SV)}$
 $= \frac{245784}{200,000 + 61446 - 13794.01}$

$= 0.99$

Modified BC ratio

$\frac{P_{u,fa}(R) - P_{u,fa}(C)}{P_{u,fa}(TI) - P_{u,fa}(SV)}$ also NAR
 $= \frac{245784 - 61446}{200,000 - 13794.01} = 0.98$

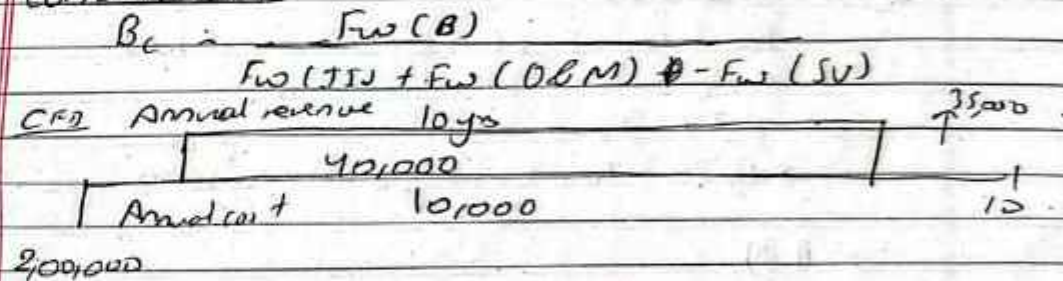
Chung

... rate - $f_{u,fa}$

Both types of ratio are less than 1 so we reject.
 (Accept if greater than 1) profit back home

(*) $F_0 \rightarrow F_{u,fa}$ multi
 And divide $\frac{1}{(1+i)^n}$

(i) $F_{u,fa}$ formulation
 Conventional



$B_c = \frac{F_{u,fa}(B)}{F_{u,fa}(TI) + F_{u,fa}(O&M) - F_{u,fa}(SV)}$

* $F_{u,fa}(B) = 40,000 \times f_{u,fa} 10\%, 10 \text{ yrs}$
 $= 40,000 \times 15.9374$
 $= 637496$

* $F_{u,fa}(TI) = 2,00,000 \times (1.1)^{10} \rightarrow \text{Ek choti mabro'rage}$
 $= 5,18,748.49$
 hursale $(1+i)^n$

suika 10% log

$$F_w(0.6M) = 10,000 \times 15.9374$$

$$= 1,59,374$$

$$F_w(SU) = 35,000$$

(Atai ma kuthe wale
ma bas ni wale)

Using conventional formula

$$B_c = \frac{F_w(0)}{F_w(II) + F_w(0.6M) - F_w(SU)}$$

$$= \frac{6,374.96}{5,18,748.79 + 1,59,374 - 35,000}$$

$$= 0.99$$

Using modified formula

$$B_c = \frac{F_w(P) - F_w(0.6M)}{F_w(II) - F_w(SU)}$$

$$= \frac{6,374.96 - 1,59,374}{5,18,748.79 - 35,000}$$

$$= 0.98$$

$$\frac{A_w}{S_u} = \frac{S_u}{F_w(II) - F_w(SU)} \rightarrow \text{future ma kunnaga}$$

$$S_u = \frac{27}{F_w(II) - F_w(SU)} \text{ uba kunnaga}$$

(x) AW

$$S_u = \frac{S_u}{F_w(II) - F_w(SU)}$$

$$F_w(II) = \frac{27}{F_w(II) - F_w(SU)}$$

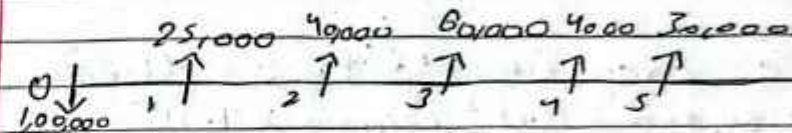
Not asked till now

Year
Cashflow
Annual return

- (*) Evaluate the following project under both types of B.C. ratio with PW and FW formulation.

Year	0	1	2	3	4	5
Cashflow	-100k	25k	40k	60k	4k	30k

MARR = 10%



MARR = 10%

SU = 0

OLM = 0

- (*) Investmental B.C. ratio with PW Formulation

$$B.C. \text{ ratio} = \frac{PW(B)}{PW(I) + PW(OLM) + PW(SU)}$$

• $PW(B) =$

$$= \frac{25,000}{(1.1)} + \frac{40,000}{(1.1)^2} + \frac{60,000}{(1.1)^3} + \frac{4,000}{(1.1)^4} + \frac{30,000}{(1.1)^5}$$

$$\text{same navako vava} = 1,46,812.18$$

divide gare present
ma lagana 1 year

$$PW(I) = 1,00,000$$

$$\therefore BC \text{ ratio} = \frac{140,812.16}{90,000} = 1.56$$

Modified also same

(*) Fw method

$$\begin{aligned} Fw(0) &= 25000 \times (1.1)^4 + 40,000 \times (1.1)^3 \\ &\quad + 60,000 \times (1.1)^2 + 70,000 \times (1.1)^1 \\ &\quad + 30,000 \times (1.1)^0 \\ &= 196,842.5 \end{aligned}$$

$$\begin{aligned} Fw(1) &= 100,000 \times (1.1)^5 \\ &= 161,051 \end{aligned}$$

$$BC \text{ ratio} = \frac{Fw(0)}{Fw(1)}$$

$$= \frac{196,842.5}{161,051}$$

$$= 1.22$$

1 sure question

(*) Comparative analysis of alternatives

Types of project

1) Independent project case

→ Suppose we have two projects independent project A and B. We can construct the following combination

Combination	Project A	Project B	Explanation
1	0	0	Do nothing
2	1	0	Select A
3	0	1	Select B
4	1	1	Select A and B

(*) Mutually exclusive project case

→ Suppose we have 3 mutually exclusive project A, B and C. Then we can construct the following combination

Combination	Project A	Project B	C	Explanation
1	0	0	0	Do nothing
2	1	0	0	Select A
3	0	1	0	Select B
4	0	0	1	Select C

J @ contingent ho

Combination	PA	PB	PC	Explanation
5	1	1	0	Select A and B
6	1	0	1	Select A and C
7	0	1	1	Select B and C

- Contingent project case
- Suppose we have three project A, B and C where A is contingent on the acceptance of both B and C, and C is contingent on both the acceptance of A. Then we can construct the following combination.

Combination	PA	PB	PC	Explanation
1	0	0	0	Do nothing
2	1	1	1	Select A, B & C
3	0	1	0	Select B
4	0	1	1	Select B and C

$$FW = -I \cdot I + S_{I \cdot I} \times A \cdot R + S_{I \cdot I}$$

Nkaile kai ana

- Govt. of Nepal has started four projects A, B, C and D. For the expansion of I.T in the country. The estimated cashflows over 10 yrs are shown in the table below. The capital investment capacity is limited to Rs 80,000 and MARR is 10% / year. Which investment according is the best by using FW formulation.

Project	A	B	C	D
I.T	32k	25k	72k	80k
A.R	7k	5k	12k	16k
S.V	4k	3k	5k	6k

- 1) Find FW of each project
- 2) Check combination
- 3) As per combination

A contingent on the acceptance of B
B D " " " " of C

Calc. of FW of all projects

FW of project A

$$FW_{10\%}(A) = -32,000 \times (1.1)^{-10} + 7,000 \times 15.937 + 4,000$$

$$= -32,000 \times 2.591 + 7,000 \times 15.937 + 4,000$$

$$= 32,562.8$$

$$FW_{10\%}(B) = -25,000 \times (1.1)^{-10} + 5,000 \times 15.937 + 3,000$$

$$= 17,843.4$$

$$FW_{10\%}(C) = -72,000 \times (1.1)^{-10} + 12,000 \times 15.937 + 5,000$$

$$= 9,799.37$$

$$\text{Tw } 10\% (D) = -80,000 \times (1.1)^{-10} + 16,000 \times 15.9374 + 6,000 \times 10.0000$$

$$= 53,499.00319$$

Construction of the combination

Combination	PA	PO	PL	Explanation
1	0	0	0	Do nothing
2	1	1	0	Accept A and B
3	0	1	0	Accept B
4	0	0	1	Accept C
5	0	0	1	Accept C and D

④ Combined projects investment and FV

Combination	Investment	FV
1	0	0
2	57,000	50,406.24
3	25,000	17,843.44
4	72,000	9499.34
5	152,000	62,998.24

Combination 5 can't be accepted due to limitation
Combination 2 could be accepted.

Decision:

If Investment Capital is limited to 80,000 only 5 is accepted. Among remaining combination combination 2 is the best because it has the highest revenue.

⑤ Capitalize Worth (CW)

- $CW \% = \frac{AW}{MARR}$
- $AW \% = \frac{-I + \frac{FV}{(1+i)^n}}{P(1+i)^n}$

Example

- Compare structure M & N given below by the given model. The structures are mutually exclusive (need to select only one project).

Items	Structure (M)	Structure (N)
I.T	12,000	40,000
SV	0	10,000
AC	2,200	1,000
Useful life	10 yrs	25 yrs
MARR	15%	

We have

$$CW\% = \frac{AW}{i}$$

Calculation of AW of structure M

$$AW(15\% M) =$$

$$AW 15\% (M) = -12000 - 2200 + 0$$

putting 15% loan

$$= \frac{-12000 - 2200}{0.15}$$

$$= -4951$$

$$CW\% (M) = \frac{AW}{i}$$

$$= \frac{-4951}{0.15} = -30,606.72$$

$$AW 15\% (N) = -40000 - 10000 + 1000$$

putting 15% 25% loan

$$= \frac{-40000 - 10000 + 1000}{0.15} = -212,793.0$$

If mutually exclusive then we
→ reject both project

$$= -16183.32 - 7141.02$$

$$CW\% (N) = \frac{AW}{i}$$

$$= \frac{-16183.32 - 7141.02}{0.15}$$

$$= -1,07,888.82 - 47,827.467$$

-47,806.67

Since, $CW\% (M) < CW\% (N)$ so we should
project M is selected.
structure.

~~~~~ Imp  
(\*) Repeatability method

Step 1: Calculation of LCM of the lives of the project.

Step 2: Calculation of PW of each project with repeatability assumption.

~~Life~~  $\frac{LCM}{Life} = \text{repetition time}$

Date \_\_\_\_\_  
Page \_\_\_\_\_

⑧ Evaluate the following two mutually exclusive projects under repeatability assumptions with PW formulation.

| Item | A        | B        |
|------|----------|----------|
| I.I  | 4,00,000 | 6,00,000 |
| A.R  | 1,20,000 | 1,60,000 |
| A.C  | 10,000   | 15,000   |
| U.L. | 6 yrs    | 8 yrs    |
| S.V. | 4000     | 7000     |

MARR = 10%

Soln

Step 1: Calculation of LCM of lives.

$$LCM: \begin{array}{r} 2 \mid 6, 8 \\ \hline 3, 4 \end{array}$$

$$= 2 \times 3 \times 4 = 24$$

Step 2: Calc of PW of the PW of the project A

Project A repeats  $\frac{24}{6} = 4$  times.

PW

Sabai value lai 0 year

Date \_\_\_\_\_  
Page \_\_\_\_\_

Calc

$$PW_{10\%}(A) = -400000 - \frac{400000}{(1.1)^6} - \frac{400000}{(1.1)^{12}} - \frac{400000}{(1.1)^{18}}$$

$$- \frac{400000}{(1.1)^{24}} + 110000 \times \text{pwIFA } 10\% \cdot 24 \text{ yrs}$$

$$= -825185.4151 +$$

$$+ \frac{4000}{(1.1)^6} + \frac{4000}{(1.1)^{12}} + \frac{4000}{(1.1)^{18}} + \frac{4000}{(1.1)^{24}}$$

$$= -825185.4151 + 1,67,789.54$$

$$PW_{10\%}(B) = -600000 - \frac{600000}{(1.1)^8} - \frac{600000}{(1.1)^{16}}$$

$$+ 1,45,000 \times \text{pwIFA } 10\% \cdot 24 \text{ yrs}$$

$$+ \frac{7000}{(1.1)^8} + \frac{7000}{(1.1)^{16}} + \frac{7000}{(1.1)^{24}}$$

$$= 297799.22$$

Since,  $PW_{10\%}(A) < PW_{10\%}(B)$  so, Structure B is selected.

1st method: - 20%  
2nd " : - 75%  
3rd " : - 5%

# (X) Risk analysis

1) Break even analysis (V.U.M):-

BEP = no loss no gain

Total Sales = total cost  $\left\{ \begin{array}{l} \text{Fixed cost} \\ \text{variable cost} \propto \text{with no. of product} \end{array} \right.$

or,  $SP \cdot x = \text{Fixed cost} + \text{Variable cost}$

or  $SP \cdot x = FC + (x \times VC \cdot pu)$   $\rightarrow$  variable cost / unit

or  $FC + (x \times VC \cdot pu)$

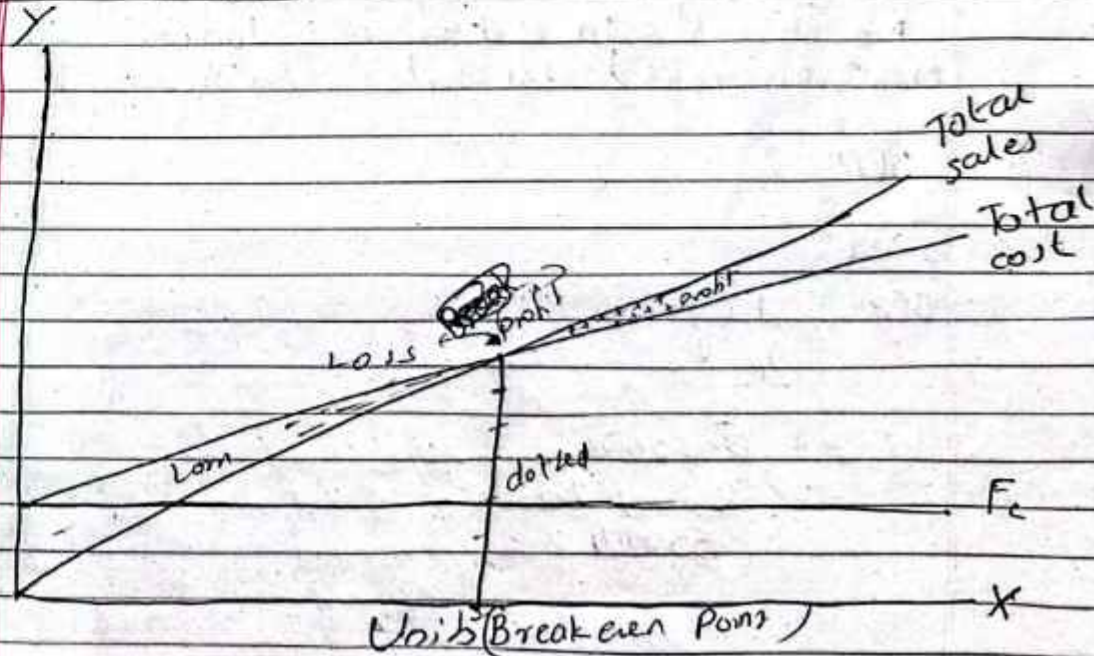
$SP(Pu) - VC(Pu)$

$x = \frac{FC}{SP(Pu) - VC(Pu)}$

$SP(Pu) - VC(Pu)$

$x = \frac{FC}{SP - VC}$

Sales and cost  
Revenue and cost



Total sales:  $Sp \times \text{no. of units}$   
(direct value)

- ⊗ Suppose Fixed cost associated with the production of the products is Rs 5,00,000. Assume that the variable cost is Rs 20,000 and the selling price is Rs 30,000 for each product and minimum sales amount is Rs 30,00,000 when sales volume is 100 units. determine the breakeven point and also plot it in BEP chart.

Sol<sup>n</sup>

Given,

Total fixed cost ( $F_c$ ) = 5,00,000

Variable cost per unit ( $V_c$ ) = 20,000

Selling price per unit ( $Sp$ ) = 30,000

No. of units sold = 100 units

(Min<sup>m</sup> sales amount) Total sales = Rs 30,00,000

BEP = ?

we have

$$\text{BEP} = \frac{F_c}{Sp - V_c}$$

$$= \frac{5,00,000}{30,000 - 20,000}$$

$$= 50 \text{ units}$$

For BEP chart

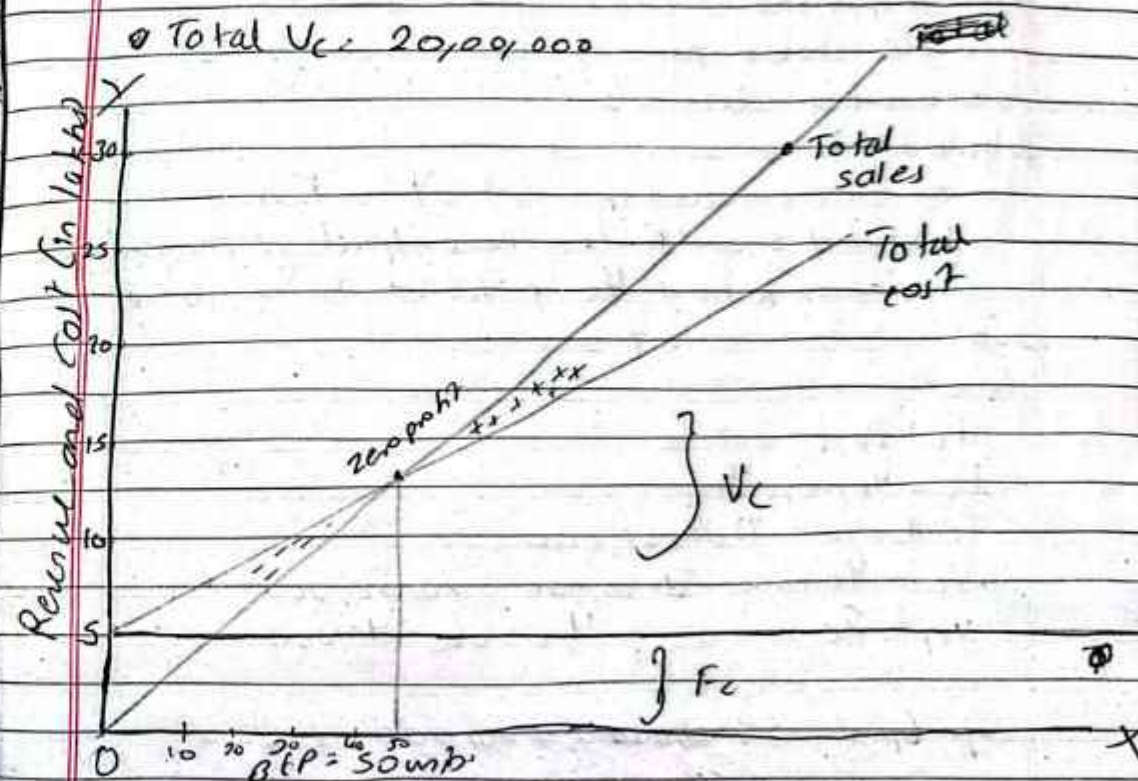
Calculation of Total cost

$$\text{Total cost} = F_c + \text{Total } V_c$$

$$= 5,00,000 + 100 \times 20,000 = 5,00,000 + 20,00,000$$

$$= 25,00,000$$

⊙ Total  $V_c$  = 20,00,000



PU > 3 asked

Date \_\_\_\_\_  
Page \_\_\_\_\_

\*) Sony company want to produce branded laptops with increasing market demand. The company has estimated following figures

$$F_c = 20,00,000$$

$$V_c = 40,00,000$$

$$\text{Total sales} = 30,00,000$$

$$\text{Sales quantity} = 50 \text{ units}$$

Required

a) Break even unit (points) of production

b) What should be the output if company desires to have the profit of Rs 5,00,000

Sol<sup>n</sup>

Given,

$$F_c = 20,00,000$$

$$V_c = 40,00,000$$

$$\text{Total sales} = 30,00,000$$

$$\text{Total } V_c = 40,00,000 - 20,00,000 = 20,00,000$$

$$V_{cp} = \frac{20,00,000}{50} = 40,000$$

$$Sp \text{ (p.u.)} = \frac{30,00,000}{50} = 60,000$$

$$\begin{aligned} \text{a) BEP} &= \frac{F_c}{Sp - V_c} = \frac{20,00,000}{60,000 - 40,000} \\ &= \frac{20,00,000}{20,000} \\ &= 100 \text{ units} \end{aligned}$$

b) ~~Let~~

$$V_c \text{ per unit} = 40,000$$

$$\text{Sales per unit} = 60,000$$

$$\text{Profit} = 5,00,000$$

$$\text{Unit to be sold} = x \text{ (assumed)}$$

$$\text{Profit} = \frac{\text{Total Sales}}{\text{Sales per unit}} - (F_c + V_c \times x)$$

$$5,00,000 = 60,000x - 20,00,000$$

$$\text{Profit} = \text{Total sales} - \text{Total cost}$$

$$5,00,000 = 60,000x - (F_c + V_c \times x)$$

$$5,00,000 = 60,000x - (20,00,000 + 40,000x)$$

$$7,00,000 = 20,000x$$

$$x = \frac{7,00,000}{20,000}$$

$$= 125 \text{ units}$$

Let 'n' be the no. of unit sale to have Rs 5,00,000 profit

we have

$$\text{Profit} = \text{T. Sales} - \text{Total cost}$$

4) old cost

Date \_\_\_\_\_  
Page \_\_\_\_\_

⊗ Breakers for dual projects

→ Consider the following two motor each of <sup>100</sup> hundred

HP output capacity.

| items   | Purchase cost | Efficiency | Life  | Maintenance cost per year | Annual tax and insurance |
|---------|---------------|------------|-------|---------------------------|--------------------------|
| Motor A | 125k          | 74%        | 10yrs | 5000                      | 1.5%                     |
| Motor B | 160k          | 92%        | 10yrs | 2500                      | 1.5%                     |

MARR = 15%

a) How many motor has to be operated at full load for the annual cost to be equal if electricity is Rs 5/(kWhr)

b) If annual operation is more than 55 hrs which motor should be selected

Sol<sup>n</sup>

Calculation of annual equivalent cost of motor

1) Cost of Calc. of CR cost

$$CR \text{ cost} = \frac{II}{PVIFA 15\% 10yrs} - \frac{SV}{FVIFA 15\% 10yrs}$$

$$= \frac{125000}{5.0188} - 0$$

$$= 24906.35$$

Note: 1 HP = 0.746 kW

Date \_\_\_\_\_  
Page \_\_\_\_\_

2) Annual maintenance cost = 5000

3) Annual tax and insurance = 1.5% of 125000  
= 1875

4) Calc of operating ~~cost~~ cost of power (electricity)

(Ghokhna pura)

Let 'X' be the no. of operation per year.

$$\text{Efficiency} = \frac{\text{output}}{\text{input}}$$

$$\text{operating expense} = \text{input} \times \text{rate} \times \text{hour}$$

$$= \frac{\text{output} \times \text{rate} \times \text{hours}}{\text{efficiency}}$$

$$= \frac{100 \times 0.746 \times 5 \times X}{0.74}$$

$$= 504.05 X$$

Total annual cost of motor A = ~~24906.35~~

= 0

$$= 24906.35 + 5000 + 1875 + 504.05 X$$

$$= 31781.35 + 504.05 X \quad \text{--- (1)}$$

Calc. of annual cost of motor B

1) Calc. of CR cost

$$\begin{aligned} \text{CR cost} &= \frac{I_i}{P_{in} A 15\% \cdot \log 2} + \frac{S_v}{E_{in} A 15\% \cdot \log 2} \\ &= \frac{160,000}{5.0188} + \frac{2400}{31810.13} \end{aligned}$$

2) Annual maintenance cost = 2500

3) Annual tax and insurance = 1.5% of 160,000  
= 2400

4) Calc. of operating cost of power (electricity)

Let 'x' be operation cost per year.

$$\begin{aligned} \text{Op. expense} &= \frac{\text{Input}}{\text{Output}} \times \text{rate} \times \text{hr} \\ &= \frac{100}{0.92} \times 0.746 \times 5 \times x \end{aligned}$$

Note: Decision criteria

- If the anticipated label is below then break even bld. select the alternative with higher variable cost
- If the anticipated label is above than break even value select the alternative with lower variable cost

$$405.43 \times$$

Eq (2)

Total annual cost of motor B

$$\begin{aligned} &= 31,880.32 + 2500 + 2400 + 405.43 \times x \\ &= 36,780.32 + 405.43 \times x \quad \text{--- Eq (2)} \end{aligned}$$

7) To get break even point, we have

Annual equivalent cost of motor A =  
" " " " " B

$$\begin{aligned} 31,781.35 + 507.05 \times x &= 36,780.32 + 405.43 \times x \\ 98.62 \times x &= 36,780.32 - 31,781.35 \\ &= 50.68 \text{ hrs} \approx 51 \text{ hrs} \end{aligned}$$

- b) If annual operation is more than 55 hrs,  
⇒ Motor B should be selected because of lower variable cost.

Parameters: (1) I (2) AC (3) AR (4) SU (5) MARR (6) Life

High chance  $\approx 75\%$

Date \_\_\_\_\_  
Page \_\_\_\_\_

## (2) Sensitivity analysis

Imp

Eg 1. Perform the sensitivity analysis — analysis of the following <sup>Project</sup> in the range of  $\pm 40\%$  in

a) I.I b) N.A.R c) SU d) useful life

$$I.I = 11,500 \quad SU = 1000$$

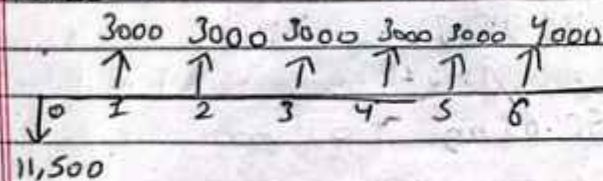
$$N.A.R = 3,000 \quad \text{Useful life} = 6 \text{ yrs}$$

$$MARR = 10\%$$

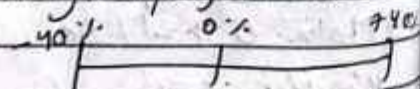
also draw a sensitivity graph and find out the most sensitive and the least sensitive parameter.

Sol<sup>n</sup>:-

C.F.D



• Calculation of  $PW$  of the given project (0% MARR)



$$PW(10\%) = -I + NAR \cdot P_{vif}(10\%, 6 \text{ yrs}) + S.V. \cdot \frac{P_{vif}(10\%, 6 \text{ yrs})}{(1+i)^n}$$

$$= -11,500 + 3,000 \times 4.3553 + 1,000 \times P_{vif} \times 10\%$$

Don't take decimal part

Date \_\_\_\_\_  
Page \_\_\_\_\_

$$= -11,500 + 3,000 \times 4.3553 + 1,000 \times 0.5645$$

$$= 2130$$

A) When initial varies  $\pm 40\%$

i)  $PW(+40\%)$

$$i) +40\% \cdot PW(10\%) = -11,500 \times 1.4 + 3,000 \times 4.3553 + 1,000 \times 0.5645$$

$$= -16,100 + 13,065.9 + 564.5$$

$$= -2,470$$

$$ii) -40\% \cdot PW(10\%) = -11,500 \times 0.6 + 13,065.9 + 564.5$$

$$= 6,730$$

B) When net annual revenue varies  $\pm 40\%$

$$i) +40\% \cdot PW(10\%) = -11,400 + 3,000 \times 1.4 \times 4.3553 + 564.49$$

$$= -11,400 + 18,292.26 + 564.49$$

$$= 7,356$$

$$ii) -40\% \cdot PW(10\%) = -11,400 + (3,000 \times 0.6) \times 4.3553 + 564.49$$

$$= -3,096$$

When

c) Salvage value varies  $\pm 40\%$

$$i) \pm 40\% \text{ Pw } 10\% = -11,500 + 13065 \cdot 91 + 1000 \times \frac{1}{1.7715} \\ = 2356.90$$

$$ii) -40\% \text{ Pw } 10\% = -11,500 + 13065 \cdot 91 + 1000 \times 0.6 \\ \frac{1}{1.7715} \\ = 1904$$

d) When useful life varies  $\pm 40\%$

Life = 6 yrs

$$+40\% \rightarrow 6 \times 1.4 = 8.4 \text{ yrs}$$

$$-40\% \rightarrow 6 \times 0.8 = 3.6 \text{ yrs}$$

$$i) +40\% \text{ Pw } 10\% = -11,500 + 3000 \times \frac{(1+i)^n - 1}{i(1+i)^n} +$$

$$\frac{1000}{(1+i)^n}$$

$$= -11,500 + 3000 \times \frac{(1.1)^{8.4} - 1}{0.1(1.1)^{8.4}} + \frac{1000}{(1.1)^{8.4}} \\ = 5476$$

$$ii) -40\% \text{ Pw } 10\% = -11,500 + 3000 \times \frac{(1.1)^{3.6} - 1}{0.1(1.1)^{3.6}} +$$

$$+ \frac{1000}{(1.1)^{3.6}}$$

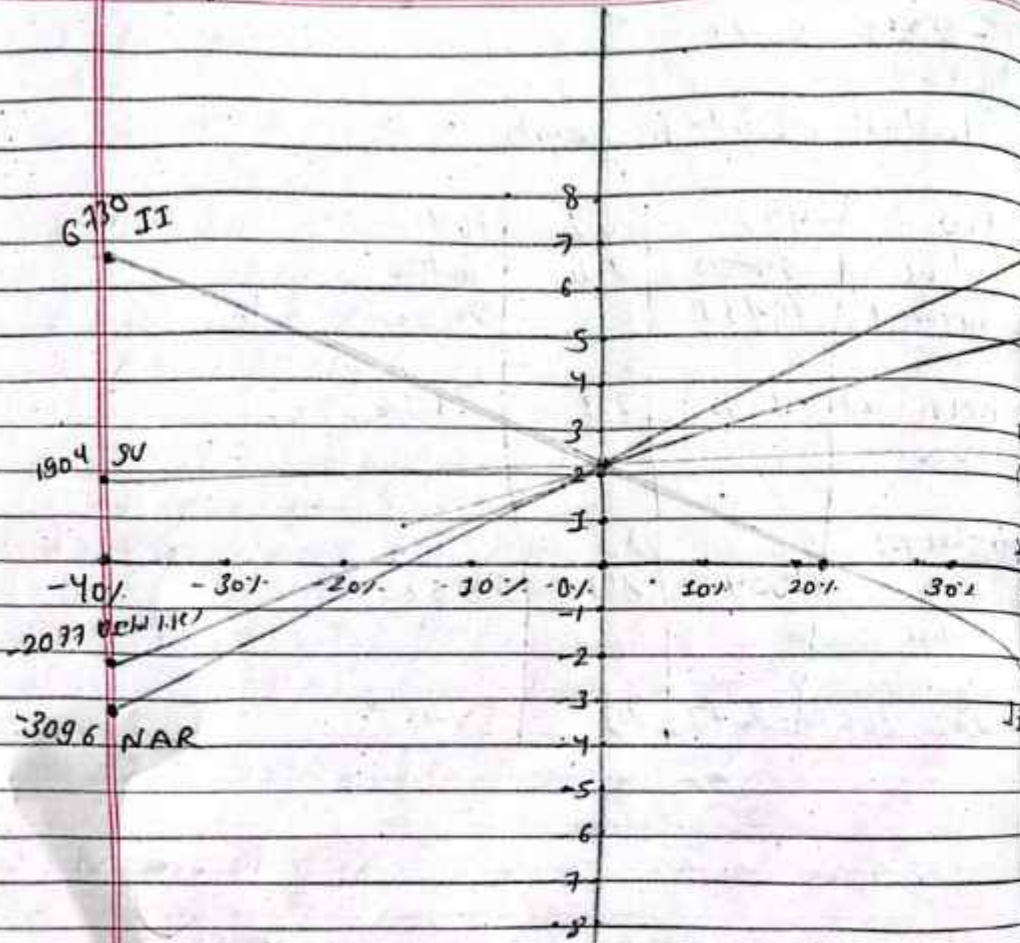
-2000 -2077

\* Calculation table for graph

| Pw                     | -40%                    | 0%   | +40%                    |
|------------------------|-------------------------|------|-------------------------|
| i) Initial investment  | <del>2470</del><br>6730 | 2130 | <del>6730</del><br>2470 |
| ii) Net Annual Revenue | -3096                   | 2130 | 7356                    |
| iii) Useful life       |                         | 2    |                         |
| ii) Salvage value      | 1904                    | 2130 | 2356                    |
| d) Useful life         | -2077                   | 2130 | 5476                    |

Exam. and graph magi

Date \_\_\_\_\_  
Page \_\_\_\_\_



Table

On the basis of calculation detail and sensitivity graph annual revenue parameter of the project is highly sensitive when compared to other parameters because AR curve is steeper than all other parameters in the diagram.

Date \_\_\_\_\_  
Page \_\_\_\_\_

Similarly, S.V. parameter is the least sensitive parameter of the project.

2-3 time cash flow

② Scenario Analysis -

→

Annual revenue  
Annual cost

(\*) From the following information of different variable or parameter and cases, calculate present worth (net present worth) for each scenario by assuming: initial investment = 1,25,000 and MARR = 15%,  $n = 5$  years

| Items                  | Worst case scenario | Most likely case scenario | Best case scenario |
|------------------------|---------------------|---------------------------|--------------------|
| units demanded         | 1600                | 2,000                     | 2,400              |
| unit price             | 48                  | 50                        | 53                 |
| variable cost per unit | 17                  | 15                        | 12                 |
| fixed cost             | 11,000              | 10,000                    | 9,000              |
| SU                     | 30,000              | 40,000                    | 50,000             |

Sol<sup>n</sup>

Calculation of annual revenue and annual cost of each scenario:

• Annual revenue:

a) worst case =  $1600 \times 48 = 76800$

b) most likely =  $2000 \times 50 = 100000$

c) Best case =  $2400 \times 53 = 1,27,200$

Annual cost (FC + VC)

a) worst case =  $11,000 + (1600 \times 17) = 38,200$

b) most likely =  $10,000 + 2000 \times 15 = 40,000$

c) Best case =  $8000 + (2400 \times 12) = 36,800$

Calculation of net-present worth

i) worst case scenario

$$\begin{aligned}
 PW(15\%) &= -1,25,000 + (76800 - 38,200) \times \text{pvifa } 15\% \text{ Sys } + 30,000 \\
 &\quad 3.3522 \quad (1.15)^5 \\
 &= 16,628.46
 \end{aligned}$$

b) most likely case

$$\begin{aligned}
 PW(15\%) &= -1,25,000 + (1,00,000 - 40,000) \times \\
 &\quad \text{pvifa } 15\% \text{ Sys } + 40,000 \\
 &\quad (1.15)^5
 \end{aligned}$$

=

c) worst case

$$\begin{aligned}
 PW(15\%) &= -1,25,000 + (1,27,200 - 36,800) \times \\
 &\quad \text{pvifa } 15\% \text{ Sys } + 50,000 \\
 &\quad (1.15)^5
 \end{aligned}$$

=

### \* Decision tree analysis:-

⇒ A glass factory that specializes in crystal is developing a substantial backlog and for this the firms mgmt each considering three courses of action. The current choice depends largely upon the future demand, which may be low, medium and high. Show the decision situation in the form of decision tree and indicate the most preferred decision and its corresponding expected value.

| Demand | probability | $S_1$<br><small>subcontracting</small> | $S_2$<br><small>being<br/>airtime</small> | $S_3$<br><small>construct<br/>facility</small> |
|--------|-------------|----------------------------------------|-------------------------------------------|------------------------------------------------|
| Low    | 0.1         | 10                                     | -20                                       | -150                                           |
| medium | 0.5         | 50                                     | 60                                        | 20                                             |
| High   | 0.4         | 50                                     | 100                                       | 200                                            |

Soln

Calc. of expected monetary value (EMV):

$$EMV(S_1) = 0.1 \times 10 + 0.5 \times 50 + 0.4 \times 50 = 46$$

$$EMV(S_2) = 0.1 \times (-20) + 0.5 \times 60 + 0.4 \times 100 = 68$$

$$EMV(S_3) = 0.1 \times (-150) + 0.5 \times 20 + 0.4 \times 200 = 75$$

